

Chapter 2: Complex Analysis

Special Functions of Mathematical Physics

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Outline

- 1 Taylor series representation
- 2 Complex variables
- 3 Derivatives
- 4 Path integrals
- 5 Taylor and Laurent series
- 6 Singularities
- 7 Residues
- 8 Residue calculus
- 9 Method of steepest descent

Where we are going

To analyze the special functions of physics we first build the tools of *complex analysis*.

- **Taylor and Laurent series** represent a function near a point, even near a singularity.
- **Analytic functions** (those with a derivative in the complex sense) obey the *Cauchy–Riemann conditions* and are infinitely differentiable.
- **Cauchy's integral theorem** and the **residue theorem** turn contour integrals into a count of poles, giving a powerful machine for evaluating real integrals and infinite sums.
- The **method of steepest descent** extracts the large-parameter behavior of integrals—the asymptotics we will need for the named functions.

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Matching a polynomial to a function I

Let

$$p_N(x) = \sum_{j=0}^N a_j x^j.$$

We choose its coefficients so that it matches a sufficiently differentiable function f at $x = 0$. For $0 \leq k \leq N$, differentiating term by term gives

$$p_N^{(k)}(x) = \sum_{j=k}^N a_j j(j-1) \cdots (j-k+1) x^{j-k} = \sum_{j=k}^N a_j \frac{j!}{(j-k)!} x^{j-k}.$$

At $x = 0$, every term with $j > k$ contains the positive power 0^{j-k} and vanishes, whereas the $j = k$ term equals $a_k k!$. Therefore

$$p_N^{(k)}(0) = k! a_k.$$

Imposing $p_N^{(k)}(0) = f^{(k)}(0)$ yields, one coefficient at a time,

$$a_k = \frac{f^{(k)}(0)}{k!}, \quad k = 0, 1, \dots, N. \quad (2.1.1)$$

For example,

$$a_0 = f(0), \quad a_1 = f'(0), \quad a_2 = \frac{f''(0)}{2!}, \quad a_3 = \frac{f^{(3)}(0)}{3!}.$$

Matching a polynomial to a function II

The resulting degree- N Taylor polynomial about 0 is

$$T_N(x) = \sum_{k=0}^N \frac{f^{(k)}(0)}{k!} x^k.$$

About an arbitrary center x_0 , apply the preceding calculation to the shifted variable $h = x - x_0$ and the function $F(h) = f(x_0 + h)$. Since $F^{(k)}(0) = f^{(k)}(x_0)$,

$$T_N(x; x_0) = \sum_{k=0}^N \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k. \quad (2.1.2)$$

If $f \in C^{N+1}$ on the interval between x_0 and x , Taylor's theorem gives

$$f(x) = T_N(x; x_0) + R_N(x), \quad R_N(x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} (x - x_0)^{N+1}$$

for some ξ between x_0 and x .

The infinite expression

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \quad (2.1.3)$$

Matching a polynomial to a function III

is initially only the *formal Taylor series*. It represents f at x precisely when $R_N(x) \rightarrow 0$ as $N \rightarrow \infty$. A real function having such a representation on a neighborhood of x_0 is called *real analytic* there. Infinite differentiability alone does not guarantee analyticity; a later example is the flat function e^{-1/x^2} with its value at 0 defined to be 0.

Examples: the elementary power series I

Exponential. The function $f(x) = e^x$ satisfies $f^{(k)}(x) = e^x$ for every $k \geq 0$, so $f^{(k)}(0) = 1$. Substitution in (2.1.3) gives

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots. \quad (2.1.4)$$

For the absolute values of consecutive terms,

$$\frac{|x|^{k+1}/(k+1)!}{|x|^k/k!} = \frac{|x|}{k+1} \rightarrow 0.$$

The ratio test therefore proves convergence for every real or complex x .

Cosine. Successive derivatives cycle as

$$\cos x, -\sin x, -\cos x, \sin x, \cos x, \dots$$

At $x = 0$ this gives the coefficient sequence $1, 0, -1, 0, 1, 0, -1, \dots$, hence

$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots. \quad (2.1.5)$$

Sine. Successive derivatives cycle as

$$\sin x, \cos x, -\sin x, -\cos x, \sin x, \dots$$

Examples: the elementary power series II

At $x = 0$ this gives $0, 1, 0, -1, 0, 1, \dots$, hence

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad (2.1.6)$$

Both series converge for all x , by the ratio test or by comparison with the absolutely convergent exponential series.

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Imaginary unit, Cartesian and polar form I

Define the imaginary unit i by $i^2 = -1$; the other square root of -1 is $-i$. A complex number has Cartesian form

$$z = x + iy, \quad x = \operatorname{Re} z, \quad y = \operatorname{Im} z,$$

where $x, y \in \mathbb{R}$. The powers of i repeat with period four:

$$i^{4q} = 1, \quad i^{4q+1} = i, \quad i^{4q+2} = -1, \quad i^{4q+3} = -i.$$

Substitute $i\theta$ into the absolutely convergent exponential series and separate even and odd powers:

$$\begin{aligned} e^{i\theta} &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \\ &= \sum_{k=0}^{\infty} \frac{i^{2k}\theta^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{i^{2k+1}\theta^{2k+1}}{(2k+1)!} \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k+1}}{(2k+1)!}. \end{aligned}$$

Comparison with (2.1.5)–(2.1.6) gives Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta. \tag{2.2.1}$$

Imaginary unit, Cartesian and polar form II

For $z \neq 0$, set

$$r = |z| = \sqrt{x^2 + y^2} > 0.$$

Choose any angle θ satisfying $x = r \cos \theta$ and $y = r \sin \theta$. Then

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}. \quad (2.2.2)$$

The angle is not unique: if θ is one argument, then every argument is $\theta + 2\pi k$, $k \in \mathbb{Z}$. The principal argument is commonly chosen as $\text{Arg } z \in (-\pi, \pi]$ (other endpoint conventions are also used). The number $z = 0$ has modulus 0 but no defined argument.

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, then

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)},$$

so moduli multiply and arguments add modulo 2π .

The complex conjugate is

$$\bar{z} = x - iy = re^{-i\theta}.$$

Consequently

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = r^2 = |z|^2,$$

and adding or subtracting z and $\bar{z}$ isolates the Cartesian components:

$$\text{Re } z = \frac{z + \bar{z}}{2}, \quad \text{Im } z = \frac{z - \bar{z}}{2i}. \quad (2.2.3)$$

Imaginary unit, Cartesian and polar form III

Replacing θ by $-\theta$ in Euler's formula gives $e^{-i\theta} = \cos \theta - i \sin \theta$. Hence

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta,$$

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta,$$

and therefore

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (2.2.4)$$

Multivalued functions and branch cuts I

An inverse relation need not define a single-valued function. For example, $y = x^2$ gives $x = \pm\sqrt{y}$ for $y > 0$. A *branch* selects one value continuously on a chosen domain; the usual real principal square root selects $+\sqrt{y}$ for $y \geq 0$.

Likewise, if $w = \sin x$, then every solution can be written

$$x = (-1)^k \arcsin w + k\pi, \quad k \in \mathbb{Z},$$

where the principal real branch has $\arcsin w \in [-\pi/2, \pi/2]$ for $w \in [-1, 1]$. For $w = 0$, this formula gives $x = k\pi$.

The complex logarithm. Let $z \neq 0$ and write $z = re^{i\theta}$ with $r > 0$. To solve $e^w = z$, write $w = u + iv$. Euler's formula gives

$$e^w = e^{u+iv} = e^u(\cos v + i \sin v).$$

Equality with $r(\cos \theta + i \sin \theta)$ requires

$$e^u = r \implies u = \log r,$$

and

$$\cos v = \cos \theta, \quad \sin v = \sin \theta \implies v = \theta + 2\pi k, \quad k \in \mathbb{Z}.$$

Thus the complete multivalued logarithm is

$$\log z = \log |z| + i(\theta + 2\pi k), \quad k \in \mathbb{Z}. \quad (2.2.5)$$

Multivalued functions and branch cuts II

A principal branch is obtained by choosing one interval of length 2π , commonly

$$-\pi < \operatorname{Arg} z < \pi, \quad \operatorname{Log} z = \log |z| + i \operatorname{Arg} z.$$

A continuous argument cannot be defined on every closed loop around 0: after one counterclockwise turn, the continuous angle increases by 2π . To obtain a single-valued logarithm, remove a curve joining the branch points 0 and ∞ . For the principal logarithm that curve is the nonpositive real axis. The boundary values satisfy

$$\operatorname{Log}(-r + i0) = \log r + i\pi, \quad \operatorname{Log}(-r - i0) = \log r - i\pi,$$

so crossing the cut changes the logarithm by $2\pi i$.

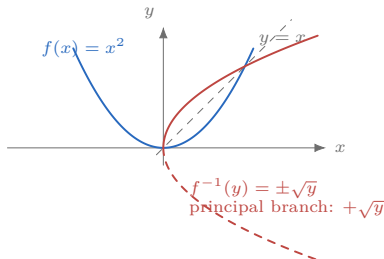
The same mechanism applies to roots. For example,

$$z^{1/2} = \sqrt{r} e^{i(\theta+2\pi k)/2} = (-1)^k \sqrt{r} e^{i\theta/2},$$

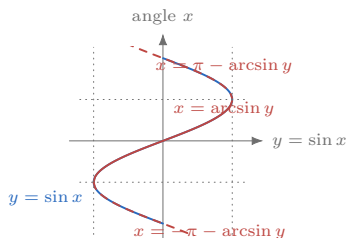
so there are two values. A branch cut permits one of them to be selected continuously.

Real graphs and inverse branches

real function and inverse branches



$\sin x$, $\arcsin$, and branch choices



principal branch uses $-\pi/2 \leq x \leq \pi/2$

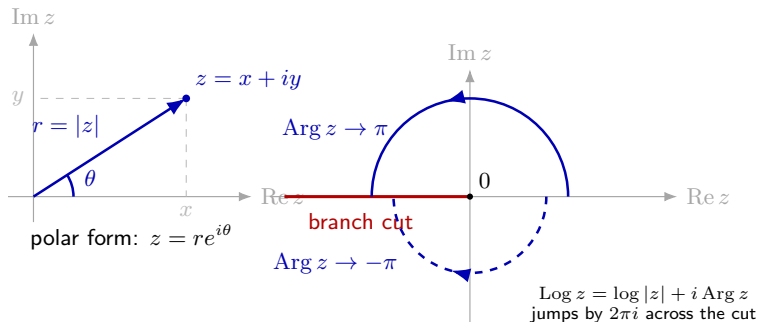
The book first motivates branch choices using real inverse functions: x^2 has two inverse branches, and $\sin x$ repeats values infinitely often. Complex roots and logarithms behave the same way, except the branch choice is made in the plane.

Complex-variable pictures and branch cuts

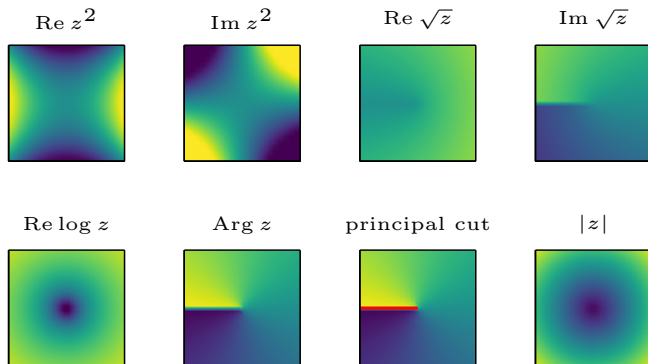
A complex-valued function has two real inputs and two real outputs:

$$z = x + iy, \quad f(z) = u(x, y) + iv(x, y).$$

Instead of one graph, we often inspect level curves, real and imaginary parts, or how a chosen branch changes around a singular point.



Component plots for complex functions



For $f(z) = u(x, y) + iv(x, y)$, a single real graph is replaced by component plots. The gallery shows the real and imaginary parts of z^2 , the principal square root, and the principal logarithm; the jump in $\operatorname{Arg} z$ is exactly the branch cut.

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The complex derivative and Cauchy–Riemann I

The complex derivative at z_0 is

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}, \quad (2.3.1)$$

provided the same limit is obtained for every complex direction of approach. A function is *holomorphic* (analytic) on an open set if this derivative exists at every point of that set; it is *entire* if the set is all of $\mathbb{C}$.

Write $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$. Along a real increment $h = \Delta x$,

$$f'(z) = u_x + iv_x.$$

Along an imaginary increment $h = i\Delta y$,

$$f'(z) = \frac{u_y + iv_y}{i} = -i(u_y + iv_y) = v_y - iu_y.$$

Equality of the real and imaginary parts gives the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x. \quad (2.3.2)$$

They are necessary for complex differentiability.

The equations are also sufficient under the standard regularity hypothesis that u, v have continuous first partial derivatives in a neighborhood. Let $h = \Delta x + i\Delta y$. Real differentiability gives

$$f(z + h) - f(z) = (u_x + iv_x)\Delta x + (u_y + iv_y)\Delta y + o(|h|).$$

The complex derivative and Cauchy–Riemann II

Using $u_y = -v_x$ and $v_y = u_x$,

$$u_y + iv_y = -v_x + iu_x = i(u_x + iv_x).$$

Therefore

$$\begin{aligned} f(z+h) - f(z) &= (u_x + iv_x)\Delta x + i(u_x + iv_x)\Delta y + o(|h|) \\ &= (u_x + iv_x)(\Delta x + i\Delta y) + o(|h|) \\ &= (u_x + iv_x)h + o(|h|). \end{aligned}$$

After division by h , the remainder satisfies $o(|h|)/h \rightarrow 0$, so

$$f'(z) = u_x + iv_x = v_y - iu_y.$$

Equivalently, the Cauchy–Riemann equations imply $f_y = if_x$.

Example: z^2 . Since

$$z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy),$$

we have

$$u_x = 2x = v_y, \quad u_y = -2y = -v_x.$$

Thus z^2 is entire and $f'(z) = 2x + i(2y) = 2z$.

Example: a branch of $\log z$. Away from 0 and the selected branch cut,

$$u = \frac{1}{2} \log(x^2 + y^2), \quad v = \text{Arg}(x + iy).$$

The complex derivative and Cauchy–Riemann III

Locally,

$$u_x = \frac{x}{x^2 + y^2}, \quad u_y = \frac{y}{x^2 + y^2}, \quad v_x = -\frac{y}{x^2 + y^2}, \quad v_y = \frac{x}{x^2 + y^2}.$$

The Cauchy–Riemann equations hold and

$$(\log z)' = u_x + iv_x = \frac{x - iy}{x^2 + y^2} = \frac{1}{z}.$$

Example: $1/z$. For $z \neq 0$,

$$\frac{\partial}{\partial x} \frac{1}{x + iy} = -\frac{1}{(x + iy)^2}, \quad \frac{\partial}{\partial y} \frac{1}{x + iy} = -\frac{i}{(x + iy)^2} = i \frac{\partial}{\partial x} \frac{1}{x + iy}.$$

Hence $1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$ and has derivative $-1/z^2$.

Counterexample: complex conjugation. For $f(z) = \bar{z} = x - iy$, $u = x$ and $v = -y$, so

$$u_x = 1 \neq -1 = v_y.$$

The Cauchy–Riemann equations fail at every point. Equivalently, $\bar{h}/h = 1$ for real h but $\bar{h}/h = -1$ for purely imaginary h ; the difference quotient therefore has no direction-independent limit.

The complex derivative and Cauchy–Riemann IV

Behavior at infinity is classified by setting $w = 1/z$. A nonconstant polynomial of degree d becomes a pole of order d at $w = 0$; e^z becomes an essential singularity there; a constant has a removable singularity at infinity.

Harmonic functions, potentials, streamlines I

Assume $f = u + iv$ is holomorphic and that the relevant second partial derivatives are continuous. Differentiate the Cauchy–Riemann equations:

$$u_x = v_y \implies u_{xx} = v_{yx},$$

$$u_y = -v_x \implies u_{yy} = -v_{xy}.$$

Equality of mixed partials, $v_{yx} = v_{xy}$, then gives

$$u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0.$$

Similarly,

$$v_x = -u_y \implies v_{xx} = -u_{yx},$$

$$v_y = u_x \implies v_{yy} = u_{xy},$$

so

$$\nabla^2 u = u_{xx} + u_{yy} = 0, \quad \nabla^2 v = v_{xx} + v_{yy} = 0. \quad (2.3.3)$$

Thus both components are harmonic. A nonconstant harmonic function cannot have a strict interior maximum or minimum (the maximum principle).

In two-dimensional potential flow, $f = \phi + i\psi$ is called the *complex potential*; $\phi = u$ is the velocity potential and $\psi = v$ is the stream function. In electrostatics, u may

Harmonic functions, potentials, streamlines II

represent the scalar potential and v its harmonic conjugate. Their level curves are orthogonal wherever their gradients are nonzero. Indeed,

$$\begin{aligned}\nabla u \cdot \nabla v &= u_x v_x + u_y v_y \\ &= v_y v_x + (-v_x) v_y = 0.\end{aligned}$$

The gradients also have equal length:

$$|\nabla u|^2 = u_x^2 + u_y^2 = v_y^2 + v_x^2 = |\nabla v|^2.$$

At a critical point both gradients may vanish, so ordinary tangent-line orthogonality is not asserted there.

Example: $f(z) = z^2$. Here

$$u = x^2 - y^2, \quad v = 2xy,$$

and the level curves are

$$x^2 - y^2 = c_1, \quad 2xy = c_2.$$

Their gradients are

$$\nabla u = (2x, -2y), \quad \nabla v = (2y, 2x),$$

so

$$\nabla u \cdot \nabla v = (2x)(2y) + (-2y)(2x) = 0.$$

Harmonic functions, potentials, streamlines III

Example: $f(z) = 1/z$. Since

$$\frac{1}{z} = \frac{x - iy}{x^2 + y^2},$$

we have

$$u = \frac{x}{x^2 + y^2}, \quad v = -\frac{y}{x^2 + y^2}.$$

For a nonzero level $u = c$,

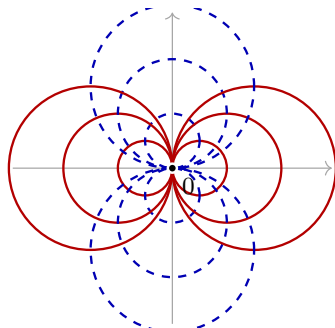
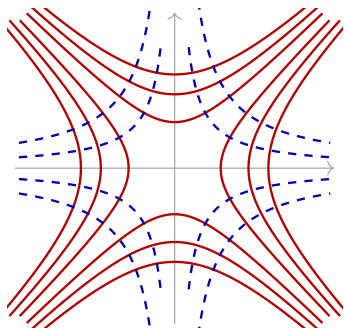
$$\begin{aligned} \frac{x}{x^2 + y^2} = c &\iff x = c(x^2 + y^2) \\ &\iff x^2 - \frac{x}{c} + y^2 = 0 \\ &\iff \left(x - \frac{1}{2c}\right)^2 + y^2 = \frac{1}{4c^2}. \end{aligned}$$

For a nonzero level $v = d$,

$$\begin{aligned} -\frac{y}{x^2 + y^2} = d &\iff -y = d(x^2 + y^2) \\ &\iff x^2 + \left(y + \frac{1}{2d}\right)^2 = \frac{1}{4d^2}. \end{aligned}$$

Thus the nonzero level curves are two orthogonal families of circles through the origin; $u = 0$ is the y -axis and $v = 0$ is the x -axis.

Harmonic functions, potentials, streamlines IV



— $u = \text{constant}$ - - - $v = \text{constant}$

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Contour integrals and a key circle integral I

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise continuously differentiable path. A partition $a = t_0 < t_1 < \cdots < t_N = b$ and sample points $\tau_j \in [t_j, t_{j+1}]$ give the Riemann sum

$$\sum_{j=0}^{N-1} f(\gamma(\tau_j))(\gamma(t_{j+1}) - \gamma(t_j)).$$

Its limit, when it exists, is the contour integral

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t))\gamma'(t) dt. \quad (2.4.1)$$

The second equality follows because $\gamma(t_{j+1}) - \gamma(t_j) \approx \gamma'(\tau_j)(t_{j+1} - t_j)$. Reversing the path reverses the sign:

$$\int_{-\gamma} f(z) dz = - \int_{\gamma} f(z) dz.$$

A closed contour may be traversed in either direction. “Positively oriented” means counterclockwise for the boundary of a bounded planar region.

Integer powers on a circle. Let $n \in \mathbb{Z}$ and parameterize the positively oriented circle $|z| = R$ by

$$z(\theta) = Re^{i\theta}, \quad dz = iRe^{i\theta} d\theta, \quad 0 \leq \theta \leq 2\pi.$$

Contour integrals and a key circle integral II

Then

$$\begin{aligned}\oint_{|z|=R} z^n dz &= \int_0^{2\pi} R^n e^{in\theta} (iRe^{i\theta}) d\theta \\ &= iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta.\end{aligned}$$

If $n = -1$, the exponential is 1, so

$$\oint_{|z|=R} \frac{dz}{z} = i \int_0^{2\pi} d\theta = 2\pi i.$$

If $n \neq -1$, then

$$\begin{aligned}iR^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta &= iR^{n+1} \left[\frac{e^{i(n+1)\theta}}{i(n+1)} \right]_0^{2\pi} \\ &= \frac{R^{n+1}}{n+1} (e^{2\pi i(n+1)} - 1) = 0,\end{aligned}$$

because $n+1$ is an integer. Translating the circle gives

$$\oint_{|z-z_0|=R} (z-z_0)^n dz = 2\pi i \delta_{n,-1}. \quad (2.4.2)$$

Contour integrals and a key circle integral III

For a general closed path avoiding z_0 ,

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z - z_0} = \text{Ind}(\gamma, z_0) \in \mathbb{Z}$$

is the winding number of γ about z_0 . To see this directly, choose a continuous polar representation along the parameter interval,

$$\gamma(t) - z_0 = r(t)e^{i\theta(t)}, \quad r(t) > 0.$$

Differentiation gives

$$\frac{\gamma'(t)}{\gamma(t) - z_0} = \frac{r'(t)}{r(t)} + i\theta'(t).$$

Therefore

$$\begin{aligned} \oint_{\gamma} \frac{dz}{z - z_0} &= \int_a^b \left(\frac{r'(t)}{r(t)} + i\theta'(t) \right) dt \\ &= [\log r(t) + i\theta(t)]_a^b. \end{aligned}$$

Because the path is closed, $r(b) = r(a)$, while $\theta(b) - \theta(a) = 2\pi \text{Ind}(\gamma, z_0)$. Hence the integral is $2\pi i$ times the winding number.

Contour integrals and a key circle integral IV

Antiderivatives. Suppose F is single-valued and holomorphic on a domain containing the path and $F' = f$. By the ordinary chain rule,

$$\frac{d}{dt} F(\gamma(t)) = F'(\gamma(t))\gamma'(t) = f(\gamma(t))\gamma'(t).$$

Thus

$$\begin{aligned}\int_{\gamma} f(z) dz &= \int_a^b \frac{d}{dt} F(\gamma(t)) dt \\ &= F(\gamma(b)) - F(\gamma(a)).\end{aligned}$$

In particular, the integral is zero on a closed path.

For integer $n \neq -1$, the single-valued primitive $F(z) = z^{n+1}/(n+1)$ exists on $\mathbb{C} \setminus \{0\}$ when needed, and therefore every closed-path integral of z^n that avoids 0 is zero. For $n = -1$, no single-valued primitive exists on a punctured neighborhood enclosing 0: the continuous logarithm along a loop changes by $2\pi i$ after one positive circuit. This is exactly the nonzero integral computed above.

Cauchy's integral theorem and formula I

Cauchy's integral theorem

Let D be a simply connected domain, let f be holomorphic on D , and let γ be a closed piecewise smooth contour in D . Then

$$\oint_{\gamma} f(z) dz = 0.$$

For the Green-theorem proof, assume additionally that $f = u + iv$ has continuous first partials and that $\gamma = \partial\Omega$ is a positively oriented simple closed curve whose interior lies in D . Since $dz = dx + i dy$,

$$\begin{aligned} f dz &= (u + iv)(dx + i dy) \\ &= (u dx - v dy) + i(v dx + u dy). \end{aligned}$$

Green's theorem gives

$$\begin{aligned} \oint_{\gamma} (u dx - v dy) &= \iint_{\Omega} ((-v)_x - u_y) dx dy = \iint_{\Omega} (-v_x - u_y) dx dy, \\ \oint_{\gamma} (v dx + u dy) &= \iint_{\Omega} (u_x - v_y) dx dy. \end{aligned}$$

Cauchy's integral theorem and formula II

The Cauchy–Riemann equations give $-v_x - u_y = 0$ and $u_x - v_y = 0$, so both real and imaginary parts vanish. The Cauchy–Goursat theorem proves the same result without the extra continuity assumption on the first partials.

Cauchy's integral formula

Let γ be a positively oriented simple closed contour, let f be holomorphic on an open set containing γ and its interior, and let z_0 lie inside γ . Then

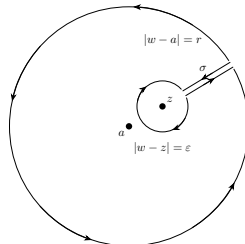
$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz. \quad (2.4.3)$$

Choose a small circle $C_\varepsilon : |z - z_0| = \varepsilon$ inside γ . The function $f(z)/(z - z_0)$ is holomorphic in the region between the two curves. The positively oriented boundary of that annular region consists of γ counterclockwise and C_ε clockwise. Cauchy's theorem therefore gives

$$0 = \oint_{\gamma} \frac{f(z)}{z - z_0} dz - \oint_{C_\varepsilon, \text{CCW}} \frac{f(z)}{z - z_0} dz,$$

so the two counterclockwise integrals are equal.

Cauchy's integral theorem and formula III



Parameterize the small circle by $z = z_0 + \varepsilon e^{i\theta}$:

$$\begin{aligned} \oint_{C_\varepsilon} \frac{f(z)}{z - z_0} dz &= \int_0^{2\pi} \frac{f(z_0 + \varepsilon e^{i\theta})}{\varepsilon e^{i\theta}} (i\varepsilon e^{i\theta}) d\theta \\ &= i \int_0^{2\pi} f(z_0 + \varepsilon e^{i\theta}) d\theta. \end{aligned}$$

Cauchy's integral theorem and formula IV

Subtract $2\pi if(z_0)$ and estimate:

$$\left| i \int_0^{2\pi} (f(z_0 + \varepsilon e^{i\theta}) - f(z_0)) d\theta \right| \leq 2\pi \max_{|z-z_0|=\varepsilon} |f(z) - f(z_0)| \\ \longrightarrow 0$$

as $\varepsilon \rightarrow 0$, by continuity of f . Hence

$$\oint_{\gamma} \frac{f(z)}{z - z_0} dz = 2\pi if(z_0),$$

which is (2.4.3).

Differentiate (2.4.3) with respect to z_0 . The contour is fixed and remains a positive distance from z_0 , so differentiation under the integral is valid. The kernel satisfies

$$\frac{\partial}{\partial z_0} (z - z_0)^{-1} = (z - z_0)^{-2}.$$

Inductively,

$$\frac{\partial^n}{\partial z_0^n} (z - z_0)^{-1} = \frac{n!}{(z - z_0)^{n+1}}.$$

Consequently, for $n = 0, 1, 2, \dots$,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz. \quad (2.4.4)$$

Cauchy's integral theorem and formula V

Equivalently, for $m = 1, 2, \dots$,

$$\oint_{\gamma} \frac{f(z)}{(z - z_0)^m} dz = \frac{2\pi i}{(m-1)!} f^{(m-1)}(z_0).$$

This formula also proves that every holomorphic function has derivatives of all orders.

Orientation and location check. Let

$$I = \oint_{\gamma} \frac{z^2}{z-2} dz.$$

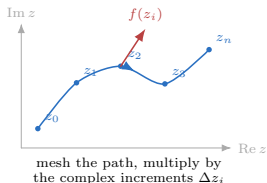
If γ winds once counterclockwise around 2, take $f(z) = z^2$ in (2.4.3):

$$I = 2\pi i f(2) = 2\pi i(2^2) = 8\pi i.$$

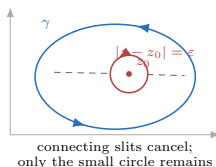
If the same geometric curve is traversed clockwise, then $I = -8\pi i$. If 2 lies outside a closed contour of winding number zero, the integrand is holomorphic throughout the enclosed region and Cauchy's theorem gives $I = 0$. More generally, the answer is $8\pi i \operatorname{Ind}(\gamma, 2)$.

Geometry behind contour arguments

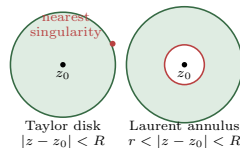
path integral as a Riemann sum



Cauchy formula: shrink to z_0



Taylor disk vs Laurent annulus



The same three pictures recur throughout the chapter: Riemann sums along a path, Cauchy's deformation to a small circle around z_0 , and the disk/annulus distinction between Taylor and Laurent convergence.

Worked example: partial fractions and Cauchy I

Evaluate

$$I = \oint_{|z|=2} \frac{e^z}{z^2(z-1)} dz,$$

with positive orientation. Write

$$\frac{1}{z^2(z-1)} = \frac{A}{z} + \frac{B}{z^2} + \frac{C}{z-1}.$$

Multiplication by $z^2(z-1)$ gives

$$\begin{aligned} 1 &= Az(z-1) + B(z-1) + Cz^2 \\ &= (A+C)z^2 + (-A+B)z - B. \end{aligned}$$

Matching the coefficients of z^2 , z , and 1 yields

$$A + C = 0, \quad -A + B = 0, \quad -B = 1.$$

Solving in order,

$$B = -1, \quad A = B = -1, \quad C = -A = 1.$$

Worked example: partial fractions and Cauchy II

Therefore

$$\begin{aligned}\frac{e^z}{z^2(z-1)} &= e^z \left(-\frac{1}{z} - \frac{1}{z^2} + \frac{1}{z-1} \right) \\ &= \frac{e^z}{z-1} - \frac{(z+1)e^z}{z^2}.\end{aligned}$$

For the simple pole at $z = 1$, Cauchy's formula gives

$$\oint_{|z|=2} \frac{e^z}{z-1} dz = 2\pi i e.$$

For the double-pole term, define

$$h(z) = (z+1)e^z.$$

Then

$$\begin{aligned}h'(z) &= (z+1)'e^z + (z+1)(e^z)' \\ &= e^z + (z+1)e^z = (z+2)e^z,\end{aligned}$$

so $h'(0) = 2$. The $m = 2$ form of Cauchy's derivative formula gives

$$\oint_{|z|=2} \frac{h(z)}{z^2} dz = 2\pi i h'(0) = 4\pi i.$$

Worked example: partial fractions and Cauchy III

Remembering the minus sign in the decomposition,

$$\begin{aligned} I &= 2\pi i e - 4\pi i \\ &= 2\pi i(e - 2). \end{aligned}$$

Both singularities, 0 and 1, lie inside $|z| = 2$; no other singularities are present.

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Taylor series from contour integration I

Taylor series theorem

If f is holomorphic on the disk $|z - z_0| < R$, then for every $|z - z_0| < R$,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

Choose a radius ρ satisfying

$$|z - z_0| < \rho < R$$

and integrate on $C_\rho : |\xi - z_0| = \rho$. Cauchy's formula gives

$$f(z) = \frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{\xi - z} d\xi.$$

Rewrite the denominator as

$$\xi - z = (\xi - z_0) - (z - z_0) = (\xi - z_0) \left(1 - \frac{z - z_0}{\xi - z_0} \right).$$

Since

$$\left| \frac{z - z_0}{\xi - z_0} \right| = \frac{|z - z_0|}{\rho} < 1,$$

Taylor series from contour integration II

the geometric series converges uniformly on the contour:

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n.$$

Uniform convergence permits term-by-term integration:

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{\xi - z} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n d\xi \\ &= \sum_{n=0}^{\infty} (z - z_0)^n \left[\frac{1}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi \right] \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n, \end{aligned}$$

where the last line uses (2.4.4).

The finite geometric identity

$$\frac{1}{1 - w} = \sum_{n=0}^N w^n + \frac{w^{N+1}}{1 - w}$$

Taylor series from contour integration III

also gives an explicit Taylor remainder:

$$R_N(z) = \frac{(z - z_0)^{N+1}}{2\pi i} \oint_{C_\rho} \frac{f(\xi)}{(\xi - z)(\xi - z_0)^{N+1}} d\xi.$$

If $M_\rho = \max_{C_\rho} |f|$, then the *ML* estimate yields

$$|R_N(z)| \leq \frac{M_\rho \rho}{\rho - |z - z_0|} \frac{|z - z_0|^{N+1}}{\rho^{N+1}} \rightarrow 0.$$

Generalized binomial series. Choose the branch of $(1 + z)^m$ that equals 1 at $z = 0$. Repeated differentiation gives

$$\frac{d^n}{dz^n} (1 + z)^m = m(m-1) \cdots (m-n+1)(1+z)^{m-n}.$$

At $z = 0$,

$$\frac{f^{(n)}(0)}{n!} = \frac{m(m-1) \cdots (m-n+1)}{n!} = \binom{m}{n}.$$

Hence

$$(1+z)^m = \sum_{n=0}^{\infty} \binom{m}{n} z^n = 1 + mz + \frac{m(m-1)}{2!} z^2 + \frac{m(m-1)(m-2)}{3!} z^3 + \cdots \quad (2.5.1)$$

Taylor series from contour integration IV

If $m \in \{0, 1, 2, \dots\}$, the factor $m - n + 1$ becomes zero and the series terminates, so the result is a polynomial valid for every z . Otherwise the nearest obstruction to the expansion about 0 is $z = -1$: it is a pole when m is a negative integer and a branch point when m is nonintegral. Therefore the radius of convergence is 1.

Analytic continuation I

A local Taylor series determines a holomorphic function uniquely on its disk of convergence. If a second Taylor disk overlaps the first, the two local formulas must agree throughout the overlap once they agree on any subset having an accumulation point. Repeating this process extends the function along chains of overlapping domains; this is analytic continuation.

For example,

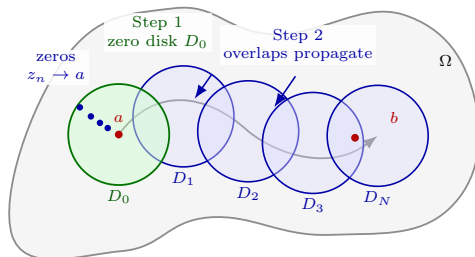
$$\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} = \sum_{n=0}^{\infty} (-1)^n z^{2n}, \quad |z| < 1.$$

The function itself is defined away from the poles $z = \pm i$, but the Taylor series about 0 stops at radius 1 because those are the nearest singularities. By contrast,

$$e^{-z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{n!}$$

is entire and has infinite radius of convergence.

Analytic continuation II



Identity theorem

Let f and g be holomorphic on a connected domain D . If $f = g$ on a set having an accumulation point in D , then $f \equiv g$ on D .

Set $h = f - g$. Suppose zeros z_k of h converge to $a \in D$. Continuity gives $h(a) = 0$. If some derivative of h at a were nonzero, let m be the smallest index with $h^{(m)}(a) \neq 0$. The Taylor series would factor as

$$h(z) = (z - a)^m \left(\frac{h^{(m)}(a)}{m!} + \frac{h^{(m+1)}(a)}{(m+1)!}(z - a) + \cdots \right) = (z - a)^m q(z),$$

Analytic continuation III

where $q(a) = h^{(m)}(a)/m! \neq 0$. Continuity would make $q(z) \neq 0$ in a small disk around a , so a would be the only zero there, contradicting the accumulating zeros. Hence all derivatives of h at a vanish and its Taylor series is zero in a neighborhood of a .

Now let

$$S = \{z \in D : h \text{ vanishes on a neighborhood of } z\}.$$

The set S is nonempty and open. It is also closed relative to D : a limit point of S is an accumulation point of zeros, so the preceding argument places it in S .

Connectedness of D implies $S = D$, hence $h \equiv 0$.

Splicing corollary. Let f_U be holomorphic on U and f_V holomorphic on V . If they agree on $U \cap V$, then

$$F(z) = \begin{cases} f_U(z), & z \in U, \\ f_V(z), & z \in V \end{cases}$$

is a well-defined holomorphic function on $U \cup V$. The identity theorem supplies uniqueness: two continuations defined on the same connected domain and agreeing near one point agree everywhere on that domain. Different continuations around branch points may return to different branches; this is the monodromy obstruction, not a failure of local uniqueness.

Laurent series I

Suppose f is holomorphic on an annulus

$$A = \{z : r < |z - z_0| < R\}.$$

Choose radii $r < \rho_1 < |z - z_0| < \rho_2 < R$. Cauchy's formula on the annular region gives, with both circles written counterclockwise,

$$f(z) = \frac{1}{2\pi i} \oint_{|\xi - z_0| = \rho_2} \frac{f(\xi)}{\xi - z} d\xi - \frac{1}{2\pi i} \oint_{|\xi - z_0| = \rho_1} \frac{f(\xi)}{\xi - z} d\xi.$$

On the outer circle, $|z - z_0| < |\xi - z_0|$, so

$$\frac{1}{\xi - z} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}.$$

On the inner circle, $|\xi - z_0| < |z - z_0|$, so

$$\begin{aligned} \frac{1}{\xi - z} &= -\frac{1}{z - z_0} \frac{1}{1 - \frac{\xi - z_0}{z - z_0}} \\ &= -\sum_{m=0}^{\infty} \frac{(\xi - z_0)^m}{(z - z_0)^{m+1}}. \end{aligned}$$

Substitution produces positive powers from the outer circle and negative powers from the inner circle.

Thus f has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-z_0)^n = \sum_{n=0}^{\infty} a_n(z-z_0)^n + \sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}, \quad (2.5.2)$$

convergent throughout the annulus. To extract a coefficient, multiply by $(z-z_0)^{-k-1}$ and integrate on any positively oriented circle γ in the annulus:

$$\begin{aligned} \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z-z_0)^{k+1}} dz &= \frac{1}{2\pi i} \sum_{n=-\infty}^{\infty} a_n \oint_{\gamma} (z-z_0)^{n-k-1} dz \\ &= \frac{1}{2\pi i} a_k (2\pi i) = a_k. \end{aligned}$$

Therefore, for every integer k ,

$$a_k = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z-z_0)^{k+1}} dz. \quad (2.5.3)$$

Term-by-term integration is justified by uniform convergence on compact subannuli.

Example: the flat real function and an essential complex singularity. Define the real function

$$F(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Laurent series III

For each $m \geq 0$,

$$\lim_{x \rightarrow 0} |x|^{-m} e^{-1/x^2} = 0.$$

Indeed, with $t = 1/x^2 \rightarrow \infty$, the expression is $t^{m/2} e^{-t}$, and the exponential dominates every power.

For $x \neq 0$, induction shows

$$F^{(n)}(x) = P_n(1/x) e^{-1/x^2}$$

for a polynomial P_n . The induction step follows from

$$\frac{d}{dx} \left[P_n(1/x) e^{-1/x^2} \right] = \left[-x^{-2} P'_n(1/x) + 2x^{-3} P_n(1/x) \right] e^{-1/x^2}.$$

Assuming $F^{(n)}(0) = 0$, the derivative at the origin is

$$F^{(n+1)}(0) = \lim_{h \rightarrow 0} \frac{F^{(n)}(h)}{h} = 0$$

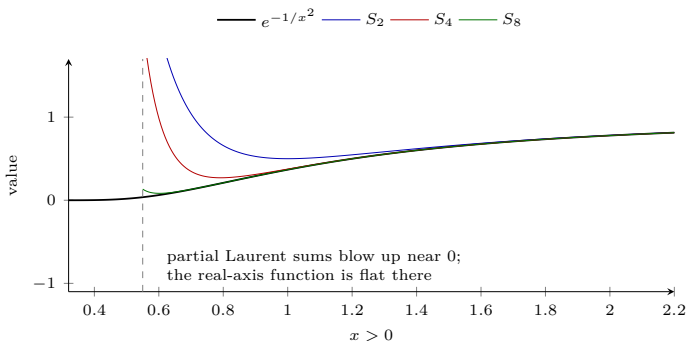
by the preceding power-exponential limit. Hence every real derivative at 0 is zero, and the real Taylor series is identically zero even though $F(x) > 0$ for $x \neq 0$.

Laurent series IV

The complex function e^{-1/z^2} cannot be made continuous at $z = 0$: along real z it tends to 0, while along $z = iy$ it equals $e^{1/y^2} \rightarrow \infty$. Its Laurent series is

$$e^{-1/z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} z^{-2n}, \quad z \neq 0,$$

which contains infinitely many negative powers; 0 is an essential singularity.



Example: sinc. Dividing the sine series by z gives, for $z \neq 0$,

$$\begin{aligned}\frac{\sin z}{z} &= \frac{1}{z} \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots \right) \\ &= 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n+1)!}.\end{aligned}$$

The right-hand side is entire and has value 1 at $z = 0$. Defining $\text{sinc } 0 = 1$ therefore removes the apparent singularity.

Laurent series and the Bernoulli numbers I

Let

$$g(z) = \frac{e^z - 1}{z} = 1 + \frac{z}{2!} + \frac{z^2}{3!} + \frac{z^3}{4!} + \frac{z^4}{5!} + \cdots = \sum_{j=0}^{\infty} b_j z^j, \quad b_j = \frac{1}{(j+1)!}.$$

Write its reciprocal as

$$\frac{1}{g(z)} = \sum_{n=0}^{\infty} a_n z^n.$$

The Cauchy product condition

$$\left(\sum_{n=0}^{\infty} a_n z^n \right) \left(\sum_{j=0}^{\infty} b_j z^j \right) = 1$$

means

$$\sum_{\ell=0}^k a_{\ell} b_{k-\ell} = \delta_{k0}.$$

Since $b_0 = 1$, the coefficients satisfy

$$a_0 = 1, \quad a_k = - \sum_{\ell=0}^{k-1} a_{\ell} b_{k-\ell} \quad (k \geq 1).$$

Laurent series and the Bernoulli numbers II

Compute the first coefficients without skipping the convolution terms:

$$a_1 = -a_0 b_1 = -\frac{1}{2},$$

$$a_2 = -(a_0 b_2 + a_1 b_1) = -\left(\frac{1}{6} - \frac{1}{4}\right) = \frac{1}{12},$$

$$\begin{aligned} a_3 &= -(a_0 b_3 + a_1 b_2 + a_2 b_1) \\ &= -\left(\frac{1}{24} - \frac{1}{12} + \frac{1}{24}\right) = 0, \end{aligned}$$

$$\begin{aligned} a_4 &= -(a_0 b_4 + a_1 b_3 + a_2 b_2 + a_3 b_1) \\ &= -\left(\frac{1}{120} - \frac{1}{48} + \frac{1}{72} + 0\right) \\ &= -\left(\frac{6 - 15 + 10}{720}\right) = -\frac{1}{720}. \end{aligned}$$

Therefore

$$\begin{aligned} \frac{z}{e^z - 1} &= 1 - \frac{z}{2} + \frac{z^2}{12} - \frac{z^4}{720} + \cdots, \\ \frac{1}{e^z - 1} &= \frac{1}{z} - \frac{1}{2} + \frac{z}{12} - \frac{z^3}{720} + \cdots. \end{aligned}$$

Laurent series and the Bernoulli numbers III

The Bernoulli numbers are defined by

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n, \quad \frac{1}{e^z - 1} = \frac{1}{z} \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n. \quad (2.5.4)$$

Thus

$$B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad B_2 = \frac{1}{6}, \quad B_3 = 0, \quad B_4 = -\frac{1}{30}, \dots$$

The nearest nonzero zeros of $e^z - 1$ are $z = \pm 2\pi i$, so the power series for $z/(e^z - 1)$ has radius 2π , and the Laurent series for $1/(e^z - 1)$ is valid on $0 < |z| < 2\pi$.

To derive the recurrence, multiply the generating function by $e^z - 1$:

$$\left(\sum_{m=1}^{\infty} \frac{z^m}{m!} \right) \left(\sum_{j=0}^{\infty} \frac{B_j}{j!} z^j \right) = z.$$

The coefficient of z^{p+1} is zero for $p \geq 1$, hence

$$\sum_{j=0}^p \frac{B_j}{j!(p+1-j)!} = 0.$$

Laurent series and the Bernoulli numbers IV

Multiplying by $(p+1)!$ gives

$$\sum_{j=0}^p \binom{p+1}{j} B_j = 0.$$

Isolating the $j = p$ term, whose coefficient is $p+1$, yields

$$B_p = -\frac{1}{p+1} \sum_{j=0}^{p-1} \binom{p+1}{j} B_j, \quad p \geq 1. \quad (2.5.5)$$

Detour: Lovelace, Bernoulli numbers, and computation I

Ada Lovelace's 1843 notes on Babbage's Analytical Engine include a tabular algorithm for Bernoulli numbers. The recurrence

$$B_p = -\frac{1}{p+1} \sum_{j=0}^{p-1} \binom{p+1}{j} B_j$$

is computational: once $B_0, \dots, B_{p-1}$ are stored, it produces B_p using only finite arithmetic. The binomial coefficients can themselves be generated by Pascal's recursion

$$\binom{q}{j} = \binom{q-1}{j-1} + \binom{q-1}{j},$$

with edge values $\binom{q}{0} = \binom{q}{q} = 1$.

The same numbers encode sums of integer powers. Define the Bernoulli polynomials by

$$\frac{te^{xt}}{e^t - 1} = \sum_{q=0}^{\infty} B_q(x) \frac{t^q}{q!}. \quad (2.5.6)$$

Detour: Lovelace, Bernoulli numbers, and computation II

Subtract the generating functions at $x + 1$ and x :

$$\begin{aligned}\sum_{q=0}^{\infty} (B_q(x+1) - B_q(x)) \frac{t^q}{q!} &= \frac{te^{(x+1)t} - te^{xt}}{e^t - 1} \\ &= te^{xt} \\ &= \sum_{q=1}^{\infty} qx^{q-1} \frac{t^q}{q!}.\end{aligned}$$

Equating coefficients gives

$$B_q(x+1) - B_q(x) = qx^{q-1}.$$

Set $q = p + 1$ and sum from $x = 1$ to $x = n$:

$$\begin{aligned}(p+1) \sum_{k=1}^n k^p &= \sum_{k=1}^n (B_{p+1}(k+1) - B_{p+1}(k)) \\ &= B_{p+1}(n+1) - B_{p+1}(1).\end{aligned}$$

Therefore

$$\sum_{k=1}^n k^p = \frac{B_{p+1}(n+1) - B_{p+1}(1)}{p+1}. \quad (2.5.7)$$

Detour: Lovelace, Bernoulli numbers, and computation III

From (2.5.6), multiplication by e^{xt} gives

$$B_q(x) = \sum_{j=0}^q \binom{q}{j} B_j x^{q-j}.$$

For $p = 1$,

$$B_2(x) = x^2 - x + \frac{1}{6},$$

so

$$\begin{aligned} \sum_{k=1}^n k &= \frac{B_2(n+1) - B_2(1)}{2} \\ &= \frac{(n+1)^2 - (n+1) + \frac{1}{6} - \frac{1}{6}}{2} \\ &= \frac{n^2 + n}{2}. \end{aligned}$$

For $p = 2$,

$$B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x,$$

Detour: Lovelace, Bernoulli numbers, and computation IV

and (2.5.7) gives

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Thus the generating function, the recurrence, and Faulhaber's polynomial sums are three forms of the same coefficient data.

The Laurent series of $\cot z$ I

Euler's formulas give

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i},$$

so

$$\cot z = i \frac{e^{iz} + e^{-iz}}{e^{iz} - e^{-iz}}. \quad (2.5.8)$$

Split the numerator and rewrite each term:

$$\begin{aligned} \cot z &= i \frac{e^{-iz}}{e^{iz} - e^{-iz}} + i \frac{e^{iz}}{e^{iz} - e^{-iz}} \\ &= i \frac{1}{e^{2iz} - 1} + i \frac{1}{1 - e^{-2iz}} \\ &= i \left[\frac{1}{e^{2iz} - 1} - \frac{1}{e^{-2iz} - 1} \right]. \end{aligned}$$

Use (2.5.4) with $w = 2iz$ and $w = -2iz$:

$$\cot z = i \sum_{p=0}^{\infty} \frac{B_p}{p!} \left[(2iz)^{p-1} - (-2iz)^{p-1} \right].$$

The Laurent series of $\cot z$ II

If p is odd, then $p - 1$ is even and the bracket is zero. Put $p = 2n$. Since $2n - 1$ is odd,

$$(-2iz)^{2n-1} = -(2iz)^{2n-1},$$

so the bracket becomes $2(2iz)^{2n-1}$. Therefore

$$\begin{aligned}\cot z &= i \sum_{n=0}^{\infty} \frac{B_{2n}}{(2n)!} 2(2iz)^{2n-1} \\ &= \sum_{n=0}^{\infty} \frac{B_{2n}}{(2n)!} \underbrace{2i(2i)^{2n-1}}_{2^{2n}i^{2n}=2^{2n}(-1)^n} z^{2n-1} \\ &= \frac{1}{z} \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} B_{2n}}{(2n)!} z^{2n}.\end{aligned}$$

Thus

$$\cot z = \frac{1}{z} - \frac{z}{3} - \frac{z^3}{45} - \frac{2z^5}{945} - \dots \quad (2.5.9)$$

For example, the z coefficient is $(-1)^1 2^2 B_2 / 2! = -4(1/6)/2 = -1/3$. The nearest other poles of $\cot z$ are at $z = \pm\pi$, so this Laurent series converges on $0 < |z| < \pi$.

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Classifying singularities I

A point z_0 is a *singular point* of f when f fails to be holomorphic there. It is an *isolated singularity* if there exists $R > 0$ such that f is holomorphic on the punctured disk

$$0 < |z - z_0| < R.$$

Only isolated singularities admit an ordinary Laurent classification. Write

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

Then:

- The singularity is **removable** if $a_n = 0$ for every $n < 0$. Defining $f(z_0) = a_0$ makes f holomorphic at z_0 . Example: $\sin z/z$ at 0.
- It is a **pole of order m** if $a_{-m} \neq 0$ and $a_n = 0$ for every $n < -m$. Equivalently,

$$f(z) = \frac{h(z)}{(z - z_0)^m}, \quad h(z_0) \neq 0,$$

with h holomorphic near z_0 .

Classifying singularities II

- It is **essential** if infinitely many negative-power coefficients are nonzero. For example,

$$\sin\left(\frac{1}{z}\right) = \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - \cdots$$

has an essential singularity at 0.

A function holomorphic except for isolated poles is *meromorphic*. For example, $1/\sin z$ is meromorphic, with simple poles at $z = n\pi$, $n \in \mathbb{Z}$.

A branch point is not an isolated singularity. For example, analytic continuation of $\log z$ once around 0 changes its value by $2\pi i$, so no single-valued branch is holomorphic on a complete punctured disk around 0. The removable/pole/essential trichotomy therefore does not apply to the branch point itself.

A point p is a boundary point of a set S if every disk centered at p intersects both S and its complement. An open set contains none of its boundary points; a closed set contains all of them. These statements concern finite points of the plane. To discuss infinity as an ordinary point, one passes to the Riemann sphere.

Let

$$X^2 + Y^2 + Z^2 = 1$$

Classifying singularities III

be the unit sphere. Stereographic projection from the north pole onto the equatorial plane identifies

$$z = \frac{X + iY}{1 - Z}. \quad (2.6.1)$$

If ϑ is angular distance from the south pole, then

$$X + iY = \sin \vartheta e^{i\phi}, \quad Z = -\cos \vartheta,$$

and therefore

$$\begin{aligned} z &= \frac{\sin \vartheta}{1 + \cos \vartheta} e^{i\phi} \\ &= \tan\left(\frac{\vartheta}{2}\right) e^{i\phi}. \end{aligned}$$

The south pole maps to 0, the equator to $|z| = 1$, and the north pole to the added point ∞ . Thus $|z| < 1$ corresponds to the southern hemisphere. A neighborhood of ∞ corresponds to $|z| > R$ for sufficiently large R , and generalized circles on the sphere project to circles or straight lines in the plane.

The nature of f at infinity is the nature of

$$F(w) = f(1/w)$$

at $w = 0$. For example, a degree- m polynomial has a pole of order m at infinity, whereas e^z has an essential singularity there.

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The residue theorem I

For an isolated singularity z_0 , the residue is the Laurent coefficient

$$\operatorname{Res}(f, z_0) = a_{-1}.$$

If a positively oriented simple contour C lies in a Laurent annulus and winds once about z_0 , term-by-term integration gives

$$\begin{aligned}\oint_C f(z) dz &= \sum_{n=-\infty}^{\infty} a_n \oint_C (z - z_0)^n dz \\ &= a_{-1}(2\pi i) \\ &= 2\pi i \operatorname{Res}(f, z_0).\end{aligned}$$

Thus

$$\oint_C f(z) dz = 2\pi i \operatorname{Res}(f, z_0). \quad (2.7.1)$$

For winding number q , the right-hand side is multiplied by q .

The residue theorem II

Residue theorem

Let f be holomorphic inside and on a positively oriented simple closed contour γ , except for finitely many isolated singularities $z_1, \dots, z_N$ in the interior. Then

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^N \operatorname{Res}(f, z_k). \quad (2.7.2)$$

To prove it, remove disjoint disks D_k centered at the singularities. On the remaining multiply connected region, f is holomorphic. Its positively oriented boundary consists of γ counterclockwise and each small circle clockwise, hence

$$0 = \oint_{\gamma} f(z) dz - \sum_{k=1}^N \oint_{\partial D_k, \text{CCW}} f(z) dz.$$

Rearranging and shrinking the circles gives

$$\oint_{\gamma} f(z) dz = \sum_{k=1}^N 2\pi i \operatorname{Res}(f, z_k),$$

which proves the theorem. For a non-simple contour, each residue is multiplied by the corresponding winding number.

The residue theorem III

Residue at a pole of order m . Write

$$h(z) = (z - z_0)^m f(z),$$

where h is holomorphic and $h(z_0) \neq 0$. Taylor expansion gives

$$h(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(z_0)}{j!} (z - z_0)^j.$$

Dividing by $(z - z_0)^m$,

$$f(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(z_0)}{j!} (z - z_0)^{j-m}.$$

The exponent equals -1 exactly when $j = m - 1$. Therefore

$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \left. \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] \right|_{z=z_0}. \quad (2.7.3)$$

For $m = 1$ this reduces to

$$\operatorname{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z),$$

and for $m = 2$ it becomes the first derivative of $(z - z_0)^2 f(z)$ at z_0 .

The residue theorem IV

A quotient at a simple zero. Suppose p and q are holomorphic, $q(z_0) = 0$, and $q'(z_0) \neq 0$. Then

$$p(z) = p(z_0) + O(z - z_0),$$

while

$$q(z) = q'(z_0)(z - z_0) + O((z - z_0)^2) = (z - z_0)(q'(z_0) + O(z - z_0)).$$

Hence

$$\frac{p(z)}{q(z)} = \frac{1}{z - z_0} \frac{p(z_0) + O(z - z_0)}{q'(z_0) + O(z - z_0)},$$

and the coefficient of $(z - z_0)^{-1}$ is

$$\operatorname{Res}\left(\frac{p}{q}, z_0\right) = \frac{p(z_0)}{q'(z_0)}. \quad (2.7.4)$$

For example,

$$\operatorname{Res}\left(\frac{1}{z^2 + 1}, i\right) = \frac{1}{2i}.$$

The residue theorem V

For a double pole,

$$\begin{aligned}\operatorname{Res}\left(\frac{1}{(z^2+1)^2}, i\right) &= \left.\frac{d}{dz} \frac{1}{(z+i)^2}\right|_{z=i} \\&= \left.-\frac{2}{(z+i)^3}\right|_{z=i} \\&= -\frac{2}{(2i)^3} = -\frac{2}{-8i} = \frac{1}{4i}.\end{aligned}$$

Residue at infinity. Suppose f is holomorphic for $|z| > R$ except possibly at ∞ , and has an exterior Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n.$$

Positive orientation around the exterior region is clockwise, so

$$\operatorname{Res}(f, \infty) = \frac{1}{2\pi i} \oint_{|z|=R, \text{CW}} f(z) dz = -a_{-1}. \quad (2.7.5)$$

The residue theorem VI

With $w = 1/z$, a counterclockwise large circle in z maps to a clockwise small circle in w , and $dz = -dw/w^2$. Thus

$$\begin{aligned}\oint_{|z|=R, \text{CCW}} f(z) dz &= \oint_{|w|=1/R, \text{CW}} f(1/w) \left(-\frac{dw}{w^2}\right) \\ &= \oint_{|w|=1/R, \text{CCW}} \frac{f(1/w)}{w^2} dw.\end{aligned}$$

Consequently

$$\text{Res}(f, \infty) = -\text{Res}\left(\frac{f(1/w)}{w^2}, 0\right). \quad (2.7.6)$$

If f is meromorphic on the Riemann sphere—equivalently, if f is rational—then it has finitely many poles and

$$\text{Res}(f, \infty) + \sum_{z_k \in \mathbb{C}} \text{Res}(f, z_k) = 0. \quad (2.7.7)$$

The qualification is essential: an arbitrary meromorphic function on $\mathbb{C}$ may have infinitely many poles, and an unqualified infinite sum of residues need not converge.

Examples at infinity.

- For $f(z) = 1/z$, the finite residue is 1, so $\text{Res}(f, \infty) = -1$.

The residue theorem VII

- For $f(z) = z$, there is a pole at infinity but no z^{-1} coefficient, so its residue at infinity is 0. Pole order and residue are different data.
- For $f(z) = (z + 1)/z^2 = z^{-1} + z^{-2}$, the residue at 0 is 1 and the residue at infinity is -1 .
- For $f(z) = 1/(z^2 + 1)$, the finite residues are $1/(2i)$ at i and $-1/(2i)$ at $-i$; their sum and the residue at infinity are both zero.

Also,

$$\begin{aligned}\operatorname{Res}\left(\frac{\sin z}{z^2}, 0\right) &= \left.\frac{d}{dz} \sin z\right|_{z=0} = 1, \\ \operatorname{Res}(z + 1, \infty) &= -\operatorname{Res}\left(\frac{1/w + 1}{w^2}, 0\right) \\ &= -\operatorname{Res}\left(\frac{1}{w^3} + \frac{1}{w^2}, 0\right) = 0.\end{aligned}$$

Two contour integrals. On $|z| = 1$,

$$\frac{\sin z}{z^2} = \frac{z - z^3/3! + z^5/5! - \cdots}{z^2} = \frac{1}{z} - \frac{z}{3!} + \frac{z^3}{5!} - \cdots.$$

The residue theorem VIII

The residue at 0 is 1, so

$$\oint_{|z|=1} \frac{\sin z}{z^2} dz = 2\pi i.$$

On the other hand,

$$\begin{aligned} \frac{z^2}{\sin z} &= \frac{z^2}{z(1 - z^2/3! + z^4/5! - \dots)} \\ &= \frac{z}{1 - z^2/3! + z^4/5! - \dots}, \end{aligned}$$

which extends holomorphically to 0 with value 0. There are no other zeros of $\sin z$ in $|z| < 1$, so

$$\oint_{|z|=1} \frac{z^2}{\sin z} dz = 0.$$

Example: $z^2/(e^z - 1)$. The denominator vanishes when

$$e^z = 1 \iff z = 2\pi im, \quad m \in \mathbb{Z}.$$

The residue theorem IX

At $m = 0$,

$$\begin{aligned}\frac{z^2}{e^z - 1} &= \frac{z^2}{z + z^2/2 + z^3/3! + \dots} \\ &= \frac{z}{1 + z/2 + z^2/3! + \dots},\end{aligned}$$

so 0 is removable and the residue there is zero. For $m \neq 0$, the zeros of $e^z - 1$ are simple because its derivative is $e^z = 1$. Formula (2.7.4) gives

$$\begin{aligned}\operatorname{Res}\left(\frac{z^2}{e^z - 1}, 2\pi im\right) &= \frac{(2\pi im)^2}{e^{2\pi im}} \\ &= -4\pi^2 m^2.\end{aligned}$$

This function has infinitely many poles, so the finite-pole global sum (2.7.7) is not applicable.

Branch-point example and a $\sum 1/n^2$ via residues I

A branch point is not a residue. Consider

$$f(z) = \frac{z^{-k}}{z+1}, \quad 0 < k < 1,$$

with the branch $0 < \arg z < 2\pi$, cut along the positive real axis. At the ordinary simple pole $z = -1$, the branch has $\arg(-1) = \pi$, so

$$\begin{aligned} \operatorname{Res}(f, -1) &= (-1)^{-k} \\ &= (e^{i\pi})^{-k} = e^{-i\pi k}. \end{aligned}$$

At $z = 0$, however, z^{-k} changes by the nontrivial factor $e^{-2\pi i k}$ after one circuit. Thus 0 is a branch point, not an isolated singularity, and no ordinary residue is defined there.

A small circle $|z| = \varepsilon < 1$ nevertheless has a vanishing contribution. On it,

$$|f(z)| \leq \frac{\varepsilon^{-k}}{1 - \varepsilon},$$

and the circle has length $2\pi\varepsilon$. Therefore

$$\left| \int_{|z|=\varepsilon} f(z) dz \right| \leq \frac{2\pi\varepsilon^{1-k}}{1 - \varepsilon} \rightarrow 0$$

Branch-point example and a $\sum 1/n^2$ via residues II

because $1 - k > 0$. A vanishing small-loop integral does not turn a branch point into an isolated singularity.

The Basel sum. Let

$$F(z) = \frac{\pi \cot(\pi z)}{z^2}.$$

Near an integer n , write $z = n + w$. Periodicity gives $\cot(\pi(n + w)) = \cot(\pi w)$, and (2.5.9) gives

$$\pi \cot(\pi z) = \frac{1}{w} - \frac{\pi^2 w}{3} - \frac{\pi^4 w^3}{45} - \dots.$$

If $n \neq 0$, the factor $1/z^2$ is holomorphic at n with value $1/n^2$. Hence

$$\operatorname{Res}(F, n) = \frac{1}{n^2}, \quad n \neq 0.$$

At the origin,

$$\begin{aligned} F(z) &= \frac{1}{z^2} \left(\frac{1}{z} - \frac{\pi^2 z}{3} - \frac{\pi^4 z^3}{45} - \dots \right) \\ &= \frac{1}{z^3} - \frac{\pi^2}{3z} - \frac{\pi^4 z}{45} - \dots, \end{aligned}$$

so

$$\operatorname{Res}(F, 0) = -\frac{\pi^2}{3}.$$

Branch-point example and a $\sum 1/n^2$ via residues III

Let Q_N be the positively oriented square with vertices

$$\left(N + \frac{1}{2}\right)(\pm 1 \pm i).$$

It encloses the integer poles $-N, -N+1, \dots, N$, has perimeter $8(N+1/2)$, and satisfies $|z| \geq N+1/2$ on its boundary.

For $z = x + iy$,

$$|\sin(\pi z)|^2 = \sin^2(\pi x) + \sinh^2(\pi y),$$

$$|\cos(\pi z)|^2 = \cos^2(\pi x) + \sinh^2(\pi y).$$

On a vertical side, $x = \pm(N+1/2)$, so $|\cot(\pi z)| \leq 1$. On a horizontal side, $|y| = N+1/2 \geq 1/2$, so

$$|\cot(\pi z)|^2 \leq \frac{1 + \sinh^2(\pi y)}{\sinh^2(\pi y)} = \coth^2(\pi y) \leq \coth^2(\pi/2) < 4.$$

Thus $|\cot(\pi z)| < 2$ on Q_N .

Branch-point example and a $\sum 1/n^2$ via residues IV

The ML estimate now gives

$$\left| \frac{1}{2\pi i} \oint_{Q_N} F(z) dz \right| \leq \frac{1}{2\pi} \frac{2\pi}{(N+1/2)^2} 8(N+1/2) \\ = \frac{8}{N+1/2} \rightarrow 0.$$

By the residue theorem,

$$\frac{1}{2\pi i} \oint_{Q_N} F(z) dz = -\frac{\pi^2}{3} + \sum_{n=1}^N \frac{1}{n^2} + \sum_{n=-N}^{-1} \frac{1}{n^2} \\ = -\frac{\pi^2}{3} + 2 \sum_{n=1}^N \frac{1}{n^2}.$$

Taking $N \rightarrow \infty$ yields

$$0 = -\frac{\pi^2}{3} + 2 \sum_{n=1}^{\infty} \frac{1}{n^2},$$

so

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \quad (2.7.8)$$

Branch-point example and a $\sum 1/n^2$ via residues V

More generally, for positive integers s ,

$$\sum_{n=1}^{\infty} \frac{1}{n^{2s}} = \frac{(-1)^{s+1} (2\pi)^{2s} B_{2s}}{2(2s)!} = \frac{(2\pi)^{2s} |B_{2s}|}{2(2s)!}.$$

Euler's Basel shortcut for the Basel problem I

A second route starts from the Weierstrass product

$$\frac{\sin(\pi x)}{\pi x} = \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right). \quad (2.7.9)$$

The product formula itself requires proof; once it is established, coefficient comparison is legitimate because the product converges uniformly on compact subsets.

For the finite product

$$P_N(x) = \prod_{n=1}^N \left(1 - \frac{x^2}{n^2}\right),$$

the constant term is 1. The coefficient of x^2 is obtained by choosing $-x^2/n^2$ from exactly one factor and 1 from all the others:

$$P_N(x) = 1 - x^2 \sum_{n=1}^N \frac{1}{n^2} + O(x^4).$$

Passing to the convergent infinite product gives

$$\prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right) = 1 - x^2 \sum_{n=1}^{\infty} \frac{1}{n^2} + O(x^4).$$

Euler's Basel shortcut for the Basel problem II

Multiplying by πx ,

$$\sin(\pi x) = \pi x - \pi x^3 \sum_{n=1}^{\infty} \frac{1}{n^2} + O(x^5). \quad (2.7.10)$$

The Taylor series gives independently

$$\begin{aligned} \sin(\pi x) &= \pi x - \frac{(\pi x)^3}{3!} + \frac{(\pi x)^5}{5!} - \dots \\ &= \pi x - \frac{\pi^3}{6} x^3 + O(x^5). \end{aligned}$$

Equating the x^3 coefficients in this expression and (2.7.10),

$$-\pi \sum_{n=1}^{\infty} \frac{1}{n^2} = -\frac{\pi^3}{6}.$$

Division by $-\pi$ yields

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The residue proof establishes the same identity without taking the sine product as an input.

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Real integrals by a semicircular contour I

Let Γ_R consist of the real segment from $-R$ to R and the upper semicircle $C_R : z = Re^{i\theta}$, $0 \leq \theta \leq \pi$. For an integrand with no pole on the real segment,

$$\oint_{\Gamma_R} f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz. \quad (2.8.1)$$

If $M_R = \max_{z \in C_R} |f(z)|$, then the arc length is πR and

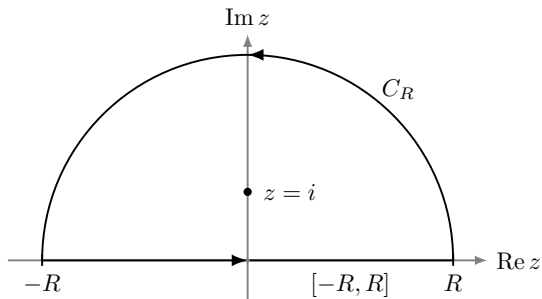
$$\left| \int_{C_R} f(z) dz \right| \leq \pi R M_R.$$

Thus the arc vanishes whenever $RM_R \rightarrow 0$. If the improper real integral converges, the residue theorem then gives

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}(f, z_k).$$

When an exponential e^{iaz} is present, its decay in one half-plane often supplies the needed estimate.

Real integrals by a semicircular contour II



Real integrals by a semicircular contour III

Example: $\int_{-\infty}^{\infty} \cos x / (1 + x^2) dx$. Consider

$$J = \int_{-\infty}^{\infty} \frac{e^{ix}}{1 + x^2} dx.$$

For $z = x + iy$ in the upper half-plane,

$$|e^{iz}| = |e^{ix-y}| = e^{-y} \leq 1.$$

On $|z| = R > 1$,

$$|1 + z^2| \geq ||z|^2 - 1| = R^2 - 1,$$

so

$$\left| \int_{C_R} \frac{e^{iz}}{1 + z^2} dz \right| \leq \frac{\pi R}{R^2 - 1} \rightarrow 0.$$

Only the pole $z = i$ lies in the upper half-plane. Since $1 + z^2 = (z - i)(z + i)$,

$$\begin{aligned} \operatorname{Res}\left(\frac{e^{iz}}{1 + z^2}, i\right) &= \frac{e^{iz}}{z + i} \Big|_{z=i} \\ &= \frac{e^{-1}}{2i}. \end{aligned}$$

Real integrals by a semicircular contour IV

Hence

$$J = 2\pi i \frac{e^{-1}}{2i} = \frac{\pi}{e}.$$

Taking real parts gives

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} dx = \frac{\pi}{e}. \quad (2.8.2)$$

The imaginary part vanishes consistently because $\sin x/(1+x^2)$ is odd.

Example: $\int_{-\infty}^{\infty} dx/(1+x^4)$. On the upper arc,

$$\left| \int_{C_R} \frac{dz}{1+z^4} \right| \leq \frac{\pi R}{R^4-1} \rightarrow 0.$$

The roots of $1+z^4=0$ are

$$z_k = e^{i(2k+1)\pi/4}, \quad k = 0, 1, 2, 3.$$

The upper-half-plane roots are

$$z_1 = e^{i\pi/4}, \quad z_2 = e^{i3\pi/4}.$$

Real integrals by a semicircular contour V

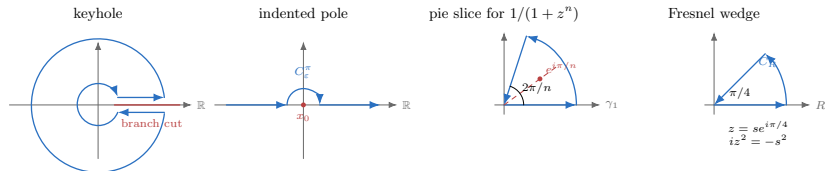
They are simple, and the derivative of $1 + z^4$ is $4z^3$, so

$$\begin{aligned}\sum \operatorname{Res}\left(\frac{1}{1+z^4}, z_k\right) &= \frac{1}{4}e^{-i3\pi/4} + \frac{1}{4}e^{-i9\pi/4} \\ &= \frac{1}{4}\left(\frac{-1-i}{\sqrt{2}} + \frac{1-i}{\sqrt{2}}\right) \\ &= -\frac{i}{2\sqrt{2}}.\end{aligned}$$

Therefore

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{dx}{1+x^4} &= 2\pi i \left(-\frac{i}{2\sqrt{2}}\right) \\ &= \frac{\pi}{\sqrt{2}}.\end{aligned}$$

Contour atlas used below



The keyhole contour compares the two sides of a branch cut; the indented contour isolates a real-axis pole; the pie-slice contour matches a rotated ray to the real-axis integral; the Fresnel wedge rotates e^{iz^2} into a Gaussian.

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ I

Let

$$I(a) = \int_0^\infty \frac{x^a}{(1+x)^2} dx, \quad -1 < a < 1.$$

The endpoint conditions follow directly: near 0 the integrand behaves as x^a , which is integrable when $a > -1$; near infinity it behaves as x^{a-2} , which is integrable when $a < 1$.

For $a \neq 0$, choose

$$z^a = e^{a \operatorname{Log} z}, \quad 0 < \arg z < 2\pi,$$

with the branch cut on the positive real axis. Use a counterclockwise keyhole contour with outer radius R and inner radius ε .

On the upper side, $z = x$ and the contribution is

$$I_{\varepsilon,R}(a) = \int_\varepsilon^R \frac{x^a}{(1+x)^2} dx.$$

On the lower side, $z = xe^{2\pi i}$ and $z^a = x^a e^{2\pi ia}$. The orientation is from R back to ε , so the contribution is

$$\int_R^\varepsilon \frac{x^a e^{2\pi ia}}{(1+x)^2} dx = -e^{2\pi ia} I_{\varepsilon,R}(a).$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ II

Thus the two straight segments sum to

$$(1 - e^{2\pi ia})I_{\varepsilon,R}(a).$$

On the outer circle, since a is real and $z^a = e^{a(\log R + i \arg z)}$, one has $|z^a| = R^a$. Hence

$$\left| \frac{z^a}{(1+z)^2} \right| \leq \frac{R^a}{(R-1)^2} = O(R^{a-2}),$$

so the arc integral is $O(R^{a-1}) \rightarrow 0$ because $a < 1$. On the inner circle,

$$\left| \frac{z^a}{(1+z)^2} \right| = O(\varepsilon^a),$$

so the arc integral is $O(\varepsilon^{a+1}) \rightarrow 0$ because $a > -1$.

The only enclosed singularity is the double pole at $z = -1$. Since

$$(z+1)^2 \frac{z^a}{(1+z)^2} = z^a,$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ III

formula (2.7.3) gives

$$\begin{aligned}\operatorname{Res}\left(\frac{z^a}{(1+z)^2}, -1\right) &= \left.\frac{d}{dz}z^a\right|_{z=-1} \\ &= a(-1)^{a-1} \\ &= ae^{i\pi(a-1)} = -ae^{i\pi a}.\end{aligned}$$

Hence the residue theorem gives, after $R \rightarrow \infty$ and $\varepsilon \rightarrow 0$,

$$(1 - e^{2\pi ia})I(a) = -2\pi iae^{i\pi a}.$$

Factor the denominator:

$$\begin{aligned}1 - e^{2\pi ia} &= e^{i\pi a}(e^{-i\pi a} - e^{i\pi a}) \\ &= -2ie^{i\pi a}\sin(\pi a).\end{aligned}$$

Therefore

$$\begin{aligned}I(a) &= \frac{-2\pi iae^{i\pi a}}{-2ie^{i\pi a}\sin(\pi a)} \\ &= \frac{\pi a}{\sin(\pi a)}.\end{aligned}$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$ IV

Thus

$$\int_0^\infty \frac{x^a}{(1+x)^2} dx = \frac{\pi a}{\sin(\pi a)}, \quad -1 < a < 1. \quad (2.8.3)$$

At $a = 0$, the right-hand side has limit 1 because $\sin(\pi a) \sim \pi a$, and directly $\int_0^\infty (1+x)^{-2} dx = 1$.

A companion integral follows from

$$K(a) = \int_0^\infty \frac{x^{-a}}{1+x} dx, \quad 0 < a < 1,$$

using the same branch. The upper and lower sides contribute

$$(1 - e^{-2\pi i a})K(a).$$

The pole at $z = -1$ is simple, with residue

$$(-1)^{-a} = e^{-i\pi a}.$$

Therefore

$$(1 - e^{-2\pi i a})K(a) = 2\pi i e^{-i\pi a}.$$

Since

$$1 - e^{-2\pi i a} = e^{-i\pi a}(e^{i\pi a} - e^{-i\pi a}) = 2ie^{-i\pi a} \sin(\pi a),$$

Keyhole contour: $\int_0^\infty x^a/(1+x)^2 dx$

we obtain

$$\int_0^\infty \frac{x^{-a}}{1+x} dx = \frac{\pi}{\sin(\pi a)}, \quad 0 < a < 1. \quad (2.8.4)$$

Cauchy principal value and a log integral I

If f has a nonintegrable singularity at an interior point x_0 , its ordinary improper integral may fail even when a symmetric cancellation exists. The Cauchy principal value is

$$\text{PV} \int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0^+} \left(\int_a^{x_0-\varepsilon} f(x) dx + \int_{x_0+\varepsilon}^b f(x) dx \right), \quad (2.8.5)$$

provided the limit exists. For example,

$$\begin{aligned} \text{PV} \int_{-1}^1 \frac{dx}{x} &= \lim_{\varepsilon \rightarrow 0^+} ([\log |x|]_{-1}^{-\varepsilon} + [\log x]_{\varepsilon}^1) \\ &= \lim_{\varepsilon \rightarrow 0^+} (\log \varepsilon - \log \varepsilon) = 0. \end{aligned}$$

The separate one-sided integrals diverge, so this is not ordinary improper convergence.

Near a simple pole,

$$f(z) = \frac{r_{-1}}{z - z_0} + O(1), \quad r_{-1} = \text{Res}(f, z_0).$$

On an arc $z = z_0 + \varepsilon e^{i\phi}$,

$$\frac{r_{-1}}{z - z_0} dz = \frac{r_{-1}}{\varepsilon e^{i\phi}} i \varepsilon e^{i\phi} d\phi = i r_{-1} d\phi.$$

Cauchy principal value and a log integral II

Thus an arc through angle θ contributes

$$\int_{C_\varepsilon^\theta} f(z) dz \longrightarrow i\sigma\theta \operatorname{Res}(f, z_0), \quad (2.8.6)$$

where $\sigma = +1$ for counterclockwise and $\sigma = -1$ for clockwise traversal.

Example. Evaluate

$$I = \int_0^\infty \frac{\log x}{x^2 - 1} dx.$$

The apparent singularity at $x = 1$ is removable, because l'Hopital's rule gives

$$\lim_{x \rightarrow 1} \frac{\log x}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{1/x}{2x} = \frac{1}{2}.$$

Thus I is an ordinary convergent improper integral.

Use

$$f(z) = \frac{\operatorname{Log} z}{z^2 - 1}$$

in the upper half-plane, where the principal logarithm has boundary values

$$\operatorname{Log}(x + i0) = \begin{cases} \log x, & x > 0, \\ \log |x| + i\pi, & x < 0. \end{cases}$$

Cauchy principal value and a log integral III

Indent clockwise above the pole $z = -1$ and above the branch point $z = 0$. The point $z = 1$ is removable and requires no indentation. Although -1 lies on the global branch cut of the principal logarithm, it is not a branch point. In a sufficiently small disk about -1 that does not contain 0 , there is a holomorphic logarithm L_{-1} satisfying $L_{-1}(-1) = i\pi$; on the upper half of that disk, L_{-1} agrees with Log . This local extension justifies treating $z = -1$ as a simple pole for the indentation calculation. The large arc is $O(\log R/R) \rightarrow 0$, and the small arc at 0 is $O(\varepsilon |\log \varepsilon|) \rightarrow 0$.

At $z = -1$,

$$\begin{aligned}\text{Res}(f, -1) &= \left. \frac{\text{Log } z}{z - 1} \right|_{z=-1} \\ &= \frac{i\pi}{-2} = -\frac{i\pi}{2}.\end{aligned}$$

The clockwise upper semicircle therefore contributes

$$\begin{aligned}-i\pi \text{Res}(f, -1) &= (-i\pi) \left(-\frac{i\pi}{2} \right) \\ &= -\frac{\pi^2}{2}.\end{aligned}$$

Cauchy principal value and a log integral IV

There are no poles inside the indented contour, so

$$0 = J - \frac{\pi^2}{2},$$

where

$$J = \text{PV} \int_{-\infty}^{\infty} \frac{\text{Log}(x + i0)}{x^2 - 1} dx.$$

Hence

$$J = \frac{\pi^2}{2}.$$

Split J at the origin. On the negative half put $x = -t$:

$$\begin{aligned} J &= \int_0^{\infty} \frac{\log t}{t^2 - 1} dt + \text{PV} \int_0^{\infty} \frac{\log t + i\pi}{t^2 - 1} dt \\ &= 2I + i\pi K, \end{aligned}$$

where

$$K = \text{PV} \int_0^{\infty} \frac{dt}{t^2 - 1}.$$

Using

$$\int \frac{dt}{t^2 - 1} = \frac{1}{2} \log \left| \frac{t-1}{t+1} \right|,$$

Cauchy principal value and a log integral V

we find

$$\begin{aligned} K &= \lim_{R \rightarrow \infty} \lim_{\varepsilon \rightarrow 0^+} \frac{1}{2} \left[\log \frac{\varepsilon}{2 - \varepsilon} + \log \frac{R - 1}{R + 1} - \log \frac{\varepsilon}{2 + \varepsilon} \right] \\ &= \lim_{\varepsilon \rightarrow 0^+} \frac{1}{2} \log \frac{2 + \varepsilon}{2 - \varepsilon} = 0. \end{aligned}$$

Therefore $J = 2I$. Since $J = \pi^2/2$,

$$\int_0^\infty \frac{\log x}{x^2 - 1} dx = \frac{\pi^2}{4}. \quad (2.8.7)$$

Worked example: a squared-log integral I

Evaluate

$$I = \int_0^{\infty} \frac{(\log x)^2}{1+x^2} dx.$$

Use the principal logarithm in the upper half-plane and

$$F(z) = \frac{\text{Log}^2 z}{1+z^2}.$$

On a large upper semicircle,

$$|F(z)| = O\left(\frac{(\log R)^2}{R^2}\right),$$

so the arc integral is $O((\log R)^2/R) \rightarrow 0$. Around 0, the arc length is $O(\varepsilon)$ and $|F(z)| = O((\log \varepsilon)^2)$, so the small-arc contribution is $O(\varepsilon(\log \varepsilon)^2) \rightarrow 0$.

Only the pole $z = i$ lies inside. Since

$$\text{Log } i = i\frac{\pi}{2},$$

Worked example: a squared-log integral II

its residue is

$$\begin{aligned}\operatorname{Res}(F, i) &= \left. \frac{\operatorname{Log}^2 z}{z + i} \right|_{z=i} \\ &= \frac{(i\pi/2)^2}{2i} \\ &= \frac{-\pi^2/4}{2i} = \frac{i\pi^2}{8}.\end{aligned}$$

The residue theorem therefore gives the upper-boundary real-axis integral

$$J := \int_{-\infty}^{\infty} \frac{\operatorname{Log}^2(x + i0)}{1 + x^2} dx = 2\pi i \frac{i\pi^2}{8} = -\frac{\pi^3}{4}. \quad (2.8.8)$$

For $x > 0$, the contribution is I . For $x < 0$, put $x = -t$ and use $\operatorname{Log}(-t + i0) = \log t + i\pi$:

$$\begin{aligned}J &= I + \int_0^{\infty} \frac{(\log t + i\pi)^2}{1 + t^2} dt \\ &= I + \int_0^{\infty} \frac{(\log t)^2 + 2i\pi \log t - \pi^2}{1 + t^2} dt \\ &= 2I + 2i\pi A - \pi^2 B,\end{aligned}$$

Worked example: a squared-log integral III

where

$$A = \int_0^{\infty} \frac{\log t}{1+t^2} dt, \quad B = \int_0^{\infty} \frac{dt}{1+t^2}.$$

For A , substitute $t = 1/u$:

$$\begin{aligned} A &= \int_{\infty}^0 \frac{-\log u}{1+u^{-2}} \left(-\frac{du}{u^2}\right) \\ &= \int_{\infty}^0 \frac{\log u}{1+u^2} du \\ &= -A. \end{aligned}$$

Hence $A = 0$. Also

$$B = [\arctan t]_0^{\infty} = \frac{\pi}{2}.$$

Thus

$$J = 2I - \frac{\pi^3}{2}.$$

Worked example: a squared-log integral IV

Equating this with (2.8.8),

$$\begin{aligned}2I - \frac{\pi^3}{2} &= -\frac{\pi^3}{4}, \\2I &= \frac{\pi^3}{4}, \\I &= \frac{\pi^3}{8}.\end{aligned}$$

Therefore

$$\int_0^\infty \frac{(\log x)^2}{1+x^2} dx = \frac{\pi^3}{8}. \quad (2.8.9)$$

Jordan's lemma and $\int \sin x/x \, dx$

Jordan-type semicircle estimate

Let C_R be the upper semicircle and let $M_R = \max_{z \in C_R} |f(z)|$. If $M_R \rightarrow 0$ as $R \rightarrow \infty$, then for every $a > 0$,

$$\int_{C_R} f(z) e^{iaz} \, dz \longrightarrow 0.$$

Analyticity of f is not needed for this arc estimate itself; it is needed elsewhere when the residue theorem is applied.

Parameterize $z = Re^{i\theta}$. Since $|e^{iaz}| = e^{-aR \sin \theta}$,

$$\begin{aligned} \left| \int_{C_R} f(z) e^{iaz} \, dz \right| &\leq M_R R \int_0^\pi e^{-aR \sin \theta} \, d\theta \\ &= 2M_R R \int_0^{\pi/2} e^{-aR \sin \theta} \, d\theta. \end{aligned}$$

Jordan's lemma and $\int \sin x/x \, dx$ II

On $0 \leq \theta \leq \pi/2$, concavity of sine gives $\sin \theta \geq 2\theta/\pi$. Therefore

$$\begin{aligned} \left| \int_{C_R} f(z) e^{iaz} \, dz \right| &\leq 2M_R R \int_0^{\pi/2} e^{-2aR\theta/\pi} \, d\theta \\ &= \frac{\pi M_R}{a} (1 - e^{-aR}) \\ &\longrightarrow 0. \end{aligned}$$

Example: the Dirichlet integral. Integrate e^{iz}/z over the upper semicircle, indenting clockwise above $z = 0$. On the large arc, take $f(z) = 1/z$, so $M_R = 1/R \rightarrow 0$ and the Jordan estimate applies. The small semicircle contributes

$$\begin{aligned} -i\pi \operatorname{Res}\left(\frac{e^{iz}}{z}, 0\right) &= -i\pi \lim_{z \rightarrow 0} e^{iz} \\ &= -i\pi. \end{aligned}$$

The indentation excludes the only pole, so the integral around the indented contour is zero. Letting the radii tend to their limits gives

$$0 = \operatorname{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} \, dx - i\pi,$$

Jordan's lemma and $\int \sin x/x \, dx$ III

and hence

$$\text{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} \, dx = i\pi. \quad (2.8.10)$$

Taking imaginary parts,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx = \pi. \quad (2.8.11)$$

The integrand $\sin x/x$ is continuous at 0 after assigning the value 1, so its integral needs no principal-value interpretation there; the principal value was required only for the auxiliary exponential integral.

Pie-slice contour: $\int_0^\infty 1/(1+x^n) dx$

Let $n > 1$ be an integer and set

$$I_n = \int_0^\infty \frac{dx}{1+x^n}.$$

Use the positively oriented boundary of the wedge $0 \leq \arg z \leq 2\pi/n$. The roots of $1+z^n=0$ are

$$z_k = e^{i(2k+1)\pi/n}, \quad k = 0, 1, \dots, n-1.$$

Exactly one, $z_0 = e^{i\pi/n}$, lies in the open wedge. It is a simple pole, and

$$\operatorname{Res}\left(\frac{1}{1+z^n}, z_0\right) = \frac{1}{nz_0^{n-1}}.$$

Because $z_0^n = -1$,

$$z_0^{n-1} = \frac{z_0^n}{z_0} = -e^{-i\pi/n},$$

so

$$\operatorname{Res}\left(\frac{1}{1+z^n}, z_0\right) = -\frac{e^{i\pi/n}}{n}.$$

The residue theorem gives

$$\oint_\Gamma \frac{dz}{1+z^n} = -\frac{2\pi i}{n} e^{i\pi/n}. \quad (2.8.12)$$

Pie-slice contour: $\int_0^\infty 1/(1+x^n) dx$ II

On the outer arc of radius $R > 1$ and angle $2\pi/n$,

$$\left| \int_{C_R} \frac{dz}{1+z^n} \right| \leq \frac{(2\pi/n)R}{R^n - 1} \rightarrow 0.$$

The outward real ray contributes I_n in the limit. On the return ray, put

$$z = re^{2\pi i/n}, \quad r: R \rightarrow 0.$$

Then

$$z^n = r^n e^{2\pi i} = r^n, \quad dz = e^{2\pi i/n} dr,$$

so the return contribution is

$$\int_R^0 \frac{e^{2\pi i/n} dr}{1+r^n} = -e^{2\pi i/n} I_n.$$

Thus (2.8.12) becomes

$$(1 - e^{2\pi i/n}) I_n = -\frac{2\pi i}{n} e^{i\pi/n}.$$

Now

$$1 - e^{2\pi i/n} = e^{i\pi/n} (e^{-i\pi/n} - e^{i\pi/n}) = -2ie^{i\pi/n} \sin(\pi/n).$$

Pie-slice contour: $\int_0^\infty 1/(1+x^n) dx$ III

After cancellation,

$$\begin{aligned} I_n &= \frac{-2\pi i e^{i\pi/n}/n}{-2i e^{i\pi/n} \sin(\pi/n)} \\ &= \frac{\pi/n}{\sin(\pi/n)}. \end{aligned}$$

Therefore

$$\int_0^\infty \frac{dx}{1+x^n} = \frac{\pi}{n} \csc\left(\frac{\pi}{n}\right), \quad n > 1. \quad (2.8.13)$$

sinc transform, Gaussian, and Fresnel integrals I

Fourier transform of sinc. Interpret the oscillatory integral as a symmetric improper integral. From (2.8.10), after replacing x by $|a|x$ and choosing the upper or lower half-plane according to the sign of a ,

$$J(a) := \text{PV} \int_{-\infty}^{\infty} \frac{e^{iat}}{t} dt = i\pi \operatorname{sgn}(a), \quad a \neq 0,$$

while $J(0) = \text{PV} \int dt/t = 0$. Since

$$\frac{\sin t}{t} e^{i\xi t} = \frac{e^{i(\xi+1)t} - e^{i(\xi-1)t}}{2it},$$

we obtain

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{\sin t}{t} e^{i\xi t} dt &= \frac{1}{2i} (J(\xi+1) - J(\xi-1)) \\ &= \frac{\pi}{2} [\operatorname{sgn}(\xi+1) - \operatorname{sgn}(\xi-1)]. \end{aligned}$$

Therefore

$$\int_{-\infty}^{\infty} \frac{\sin t}{t} e^{i\xi t} dt = \begin{cases} \pi, & |\xi| < 1, \\ \pi/2, & |\xi| = 1, \\ 0, & |\xi| > 1. \end{cases} \quad (2.8.14)$$

The half-height at $|\xi| = 1$ arises because one of the two J arguments is zero.

sinc transform, Gaussian, and Fresnel integrals II

Gaussian integral. Let

$$G = \int_{-\infty}^{\infty} e^{-x^2} dx > 0.$$

Since the integrand is nonnegative, Tonelli's theorem permits multiplication as an iterated double integral:

$$G^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy.$$

In polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, the Jacobian is r , so

$$\begin{aligned} G^2 &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta \\ &= 2\pi \left[-\frac{1}{2} e^{-r^2} \right]_0^{\infty} \\ &= 2\pi \left(\frac{1}{2} \right) = \pi. \end{aligned}$$

Because $G > 0$, $G = \sqrt{\pi}$. For $m > 0$, set $u = \sqrt{m} x$, so $dx = du/\sqrt{m}$:

$$\int_{-\infty}^{\infty} e^{-mx^2} dx = \frac{1}{\sqrt{m}} \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\frac{\pi}{m}}. \quad (2.8.15)$$

sinc transform, Gaussian, and Fresnel integrals III

Fresnel integrals. Integrate the entire function e^{iz^2} around the wedge with vertices 0, R , and $Re^{i\pi/4}$. On the circular arc $z = Re^{i\theta}$, $0 \leq \theta \leq \pi/4$,

$$|e^{iz^2}| = e^{-R^2 \sin(2\theta)}.$$

Since $\sin(2\theta) \geq 4\theta/\pi$ on this interval,

$$\begin{aligned} \left| \int_{C_R} e^{iz^2} dz \right| &\leq R \int_0^{\pi/4} e^{-R^2 \sin(2\theta)} d\theta \\ &\leq R \int_0^{\pi/4} e^{-4R^2\theta/\pi} d\theta \\ &= \frac{\pi}{4R} (1 - e^{-R^2}) \rightarrow 0. \end{aligned}$$

On the slanted ray, put $z = e^{i\pi/4}s$. Then

$$z^2 = is^2, \quad iz^2 = -s^2, \quad dz = e^{i\pi/4} ds.$$

sinc transform, Gaussian, and Fresnel integrals IV

The contour traverses this ray from $s = R$ back to $s = 0$, so Cauchy's theorem gives

$$0 = \int_0^R e^{ix^2} dx + \int_{C_R} e^{iz^2} dz - e^{i\pi/4} \int_0^R e^{-s^2} ds.$$

Letting $R \rightarrow \infty$,

$$\begin{aligned} \int_0^\infty e^{ix^2} dx &= e^{i\pi/4} \int_0^\infty e^{-s^2} ds \\ &= e^{i\pi/4} \frac{\sqrt{\pi}}{2} \\ &= \frac{\sqrt{\pi}}{2\sqrt{2}}(1 + i). \end{aligned}$$

Taking real and imaginary parts,

$$\int_0^\infty \cos(x^2) dx = \int_0^\infty \sin(x^2) dx = \frac{\sqrt{\pi}}{2\sqrt{2}}. \quad (2.8.16)$$

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The steepest-descent approximation I

Consider

$$I(s) = \int_{\Gamma} g(z) e^{sf(z)} dz, \quad s \rightarrow +\infty, \quad (2.9.1)$$

with f and g holomorphic near a contour that may be deformed without crossing singularities. Write $f = u + iv$. For a holomorphic function,

$$f'(z) = u_x + iv_x = u_x - iu_y,$$

where the second equality uses $v_x = -u_y$. Thus $f'(z_0) = 0$ implies

$$u_x(z_0) = u_y(z_0) = 0.$$

The Cauchy–Riemann equations then also give $v_x(z_0) = v_y(z_0) = 0$. Hence both the real and imaginary parts are stationary at a complex saddle.

Since u is harmonic,

$$u_{yy} = -u_{xx}.$$

Also,

$$f''(z) = u_{xx} + iv_{xx} = u_{xx} - iu_{xy}.$$

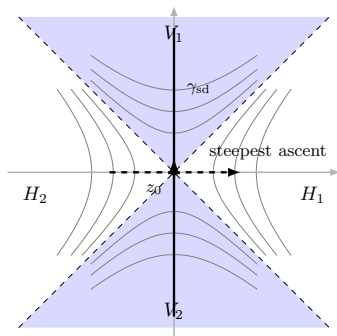
The steepest-descent approximation II

Therefore

$$D^2 u(z_0) = \begin{pmatrix} u_{xx} & u_{xy} \\ u_{xy} & -u_{xx} \end{pmatrix},$$

$$\det D^2 u(z_0) = -u_{xx}^2 - u_{xy}^2 = -|f''(z_0)|^2.$$

Thus $f''(z_0) \neq 0$ is equivalent to a nondegenerate saddle of $u = \operatorname{Re} f$.



The steepest-descent approximation III

Taylor expansion at a nondegenerate saddle gives

$$f(z) = f(z_0) + \frac{1}{2}f''(z_0)(z - z_0)^2 + O((z - z_0)^3). \quad (2.9.2)$$

Write

$$f''(z_0) = |f''(z_0)|e^{i\theta}, \quad z - z_0 = re^{i\phi}.$$

The quadratic change in the exponent is

$$\frac{1}{2}f''(z_0)(z - z_0)^2 = \frac{1}{2}|f''(z_0)|r^2e^{i(\theta+2\phi)}.$$

A steepest-descent direction makes this quantity negative real, so

$$\theta + 2\phi = \pi \pmod{2\pi}.$$

One representative is

$$\phi = \frac{\pi - \theta}{2}, \quad (2.9.3)$$

and the opposite ray gives the other half of the same descent line. Along it,

$$\frac{1}{2}f''(z_0)(z - z_0)^2 = -\frac{1}{2}|f''(z_0)|r^2.$$

The orthogonal directions satisfying $\theta + 2\phi = 0 \pmod{2\pi}$ are steepest-ascent directions.

The steepest-descent approximation IV

Assume that the contour can be deformed through z_0 along both descent rays, that this saddle dominates all other portions of the contour, and that $g(z_0) \neq 0$. Put

$$\alpha = ie^{-i\theta/2} = e^{i(\pi-\theta)/2}, \quad z = z_0 + \alpha r, \quad -\infty < r < \infty.$$

Then

$$\alpha^2 = -e^{-i\theta},$$

so

$$\begin{aligned} f''(z_0)\alpha^2 &= |f''(z_0)|e^{i\theta}(-e^{-i\theta}) \\ &= -|f''(z_0)|. \end{aligned}$$

Moreover $dz = \alpha dr$. Replacing f by its quadratic term and g by $g(z_0)$ gives

$$\begin{aligned} I(s) &\sim g(z_0)e^{sf(z_0)}\alpha \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}s|f''(z_0)|r^2\right) dr \\ &= g(z_0)e^{sf(z_0)}\alpha \sqrt{\frac{2\pi}{s|f''(z_0)|}}. \end{aligned}$$

Therefore, for the stated orientation,

$$I(s) \sim g(z_0)e^{sf(z_0)}ie^{-i\theta/2}\sqrt{\frac{2\pi}{s|f''(z_0)|}}. \quad (2.9.4)$$

The steepest-descent approximation V

Reversing the contour reverses the sign. If the saddle is an endpoint, only one Gaussian half-line contributes. If several saddles have the same dominant real part, their contributions must be summed; crossing Stokes lines can change which saddles contribute.

Worked example: large binomial coefficient I

Let $0 < m < n$, set

$$t = \frac{m}{n} \in (0, 1),$$

and assume that both m and $n - m$ are large. From

$$(1 + z)^n = \sum_{k=0}^n \binom{n}{k} z^k,$$

Cauchy's coefficient formula gives

$$\begin{aligned} \binom{n}{m} &= \frac{1}{2\pi i} \oint \frac{(1+z)^n}{z^{m+1}} dz \\ &= \oint g(z) e^{nf(z)} dz, \end{aligned}$$

where

$$g(z) = \frac{1}{2\pi i z}, \quad f(z) = \log(1+z) - t \log z.$$

The original rational-polynomial integrand is single-valued. The logarithms are used only locally after choosing branches near the positive saddle.

Differentiate:

$$f'(z) = \frac{1}{1+z} - \frac{t}{z}, \quad f''(z) = -\frac{1}{(1+z)^2} + \frac{t}{z^2}.$$

Worked example: large binomial coefficient II

The saddle equation is

$$\begin{aligned}\frac{1}{1+z_0} - \frac{t}{z_0} &= 0, \\ z_0 - t(1+z_0) &= 0, \\ (1-t)z_0 &= t,\end{aligned}$$

so

$$z_0 = \frac{t}{1-t} = \frac{m}{n-m} > 0. \quad (2.9.5)$$

At the saddle,

$$1 + z_0 = 1 + \frac{t}{1-t} = \frac{1}{1-t}.$$

Therefore

$$\begin{aligned}f(z_0) &= \log \frac{1}{1-t} - t \log \frac{t}{1-t} \\ &= -\log(1-t) - t \log t + t \log(1-t) \\ &= -t \log t - (1-t) \log(1-t).\end{aligned}$$

Worked example: large binomial coefficient III

For the second derivative,

$$\begin{aligned}f''(z_0) &= -(1-t)^2 + t \frac{(1-t)^2}{t^2} \\&= (1-t)^2 \left(-1 + \frac{1}{t}\right) \\&= (1-t)^2 \frac{1-t}{t} \\&= \frac{(1-t)^3}{t} > 0.\end{aligned}$$

Thus $\theta = 0$, and the descent line is vertical. The coefficient contour may be chosen as the circle $|z| = z_0$; at its positive real point, counterclockwise orientation has tangent $+i$, matching the phase factor in (2.9.4).

Apply (2.9.4) with $s = n$:

$$\begin{aligned}\binom{n}{m} &\sim \frac{1}{2\pi i z_0} e^{nf(z_0)} i \sqrt{\frac{2\pi}{nf''(z_0)}} \\&= \frac{1}{2\pi z_0} e^{nf(z_0)} \sqrt{\frac{2\pi}{nf''(z_0)}}.\end{aligned}$$

Worked example: large binomial coefficient IV

Now substitute

$$\begin{aligned}\frac{1}{z_0} &= \frac{1-t}{t}, & \frac{1}{f''(z_0)} &= \frac{t}{(1-t)^3} \cdot \\ \frac{1}{2\pi z_0} \sqrt{\frac{2\pi}{nf''(z_0)}} &= \frac{1-t}{2\pi t} \sqrt{\frac{2\pi t}{n(1-t)^3}} \\ &= \frac{1}{\sqrt{2\pi n}} \frac{(1-t)\sqrt{t}}{t(1-t)^{3/2}} \\ &= \frac{1}{\sqrt{2\pi nt(1-t)}}.\end{aligned}$$

Also,

$$e^{nf(z_0)} = \exp[-n(t \log t + (1-t) \log(1-t))].$$

Hence

$$\binom{n}{m} \sim \frac{\exp[-n(t \log t + (1-t) \log(1-t))]}{\sqrt{2\pi nt(1-t)}}. \quad (2.9.6)$$

Worked example: large binomial coefficient V

To rewrite the exponential in terms of m and n , use $t = m/n$ and $1 - t = (n - m)/n$:

$$e^{-nt \log t} = t^{-m} = \left(\frac{n}{m}\right)^m,$$

$$e^{-n(1-t) \log(1-t)} = (1 - t)^{-(n-m)} = \left(\frac{n}{n-m}\right)^{n-m}.$$

Thus

$$e^{nf(z_0)} = \frac{n^n}{m^m(n-m)^{n-m}}.$$

Furthermore,

$$\sqrt{2\pi nt(1-t)} = \sqrt{\frac{2\pi m(n-m)}{n}}.$$

Therefore the equivalent form is

$$\binom{n}{m} \sim \frac{1}{\sqrt{2\pi}} \frac{n^{n+1/2}}{m^{m+1/2}(n-m)^{n-m+1/2}}. \quad (2.9.7)$$

This is the result obtained by applying Stirling's approximation separately to the three factorials.

Numerical check. For $n = 88$ and $m = 18$,

$$\binom{88}{18} = 2,418,561,960,739,869,780,$$

Worked example: large binomial coefficient VI

while (2.9.6) gives

$$2.430375540 \times 10^{18}.$$

The relative error is

$$\frac{2.430375540 \times 10^{18} - 2.418561961 \times 10^{18}}{2.418561961 \times 10^{18}} \approx 4.8845 \times 10^{-3} = 0.48845\%.$$

Summary I

- Complex differentiability requires one direction-independent difference-quotient limit. With continuous first partial derivatives, this is equivalent to the Cauchy–Riemann equations; the real and imaginary parts are then harmonic.
- Cauchy's theorem implies path independence on simply connected domains. Cauchy's integral formula yields all derivatives and the local Taylor expansion, including explicit convergence and remainder estimates.
- A Laurent series is valid on an annulus. Its negative-power part classifies an isolated singularity as removable, a pole, or essential. Branch points are not isolated and do not possess ordinary residues.
- The coefficient of $(z - z_0)^{-1}$ is the residue. The residue theorem evaluates a contour integral as $2\pi i$ times the sum of enclosed residues, with winding numbers included for general contours.
- Semicircle, keyhole, indented, and wedge contours require explicit orientation and vanishing-arc estimates. They yield the Basel sum, logarithmic integrals, the Dirichlet integral, the Gaussian and Fresnel integrals, and beta-function-type formulas.

Summary II

- Steepest descent is a local Gaussian approximation around a nondegenerate dominant saddle. Its phase depends on the oriented descent direction, and multiple or endpoint saddles require modified contributions.